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# Encoded and non-genetic protein variants expand human functional proteome

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**Each stage of the Central Dogma contributes to proteome diversity through mechanisms such as heterozygosity, somatic mutations, transcriptional errors, and translational errors. As a result, a diverse array of protein variants can coexist within a single proteome, such as that of humans. However, until now, methods to detect, quantify, and evaluate the functional consequences of these variants have been lacking. Here we examined a large-scale proteogenomic dataset from 29 healthy human tissues and uncovered 13,910 confidently localized variants representing 7,215 unique single amino acid substitutions co-existing alongside their corresponding reference proteoforms [1].We found that the abundance of both genetic (SNP's, somatic mutations) and mistranslated protein variants mirrors their allele frequencies in the human population. Moreover, we show that non-genetic substitutions may provide a distinct route for exploring protein sequence space, circumventing the mutational constraints imposed by the genetic code. In addition, we provide experimental validation of non-genetic substitution on selected purified proteins. We demonstrate specific and recurring non-genetic variation patterns upon amino acid starvation in proteome-wide analyses of cancer-derived cell lines and identify hundreds of substituted non-genetic proteoforms that recur consistently in multiple healthy individuals or map to annotated protein functional sites. We propose that these substitutions constitute a novel class of functional protein phenotypic variants. Collectively, our findings indicate that non-genetic amino acid substitutions in human proteins provide an abundant source to expanding the functional proteome.**

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# Introduction

Genomes provide blueprint for the majority of proteins. However, it has become clear that the complexity of the human proteome extends beyond the canonical set of ~20,000 protein-coding genes[2]. The molecular determinants of cellular phenotype arise from an interplay between alternative splicing[3], posttranslational modifications[4], heterozygosicity[5], somatic mutations[6], microproteins and "peptideins"[7], and phenotypic variants generated by transcriptional or translational errors[8,9]. Translation errors can diversify proteomes by through amino acid substitutions[9], ribosomal frame-shifting[10] and STOP codon readthrough[11]. These processes may offer myriad ways to fine-tune cellular phenotypes.

Human heterozygosity also increases protein heterogeneity by expressing both maternal and paternal alleles. These alternative proteins may differ subtly in structure, function, stability, or solubility, resulting in various effects on cellular fitness[12]. Notably, allele-specific gene regulation can further expand the proteome's diversity [13]. For instance, cell differentiation or metabolism genes may display differential allelic expression[5]. Heterozygosity can enhance resilience against pathogenic stressors[14], mitigate the impact of deleterious homozygous mutations[15], or influence therapeutic responses[16].

Somatic mutations, also add to the encoded proteomic diversity. Although somatic mutations are frequently linked to senescence, and malignant growth, they can also provide selective advantages[17]. The foremost example is error-prone DNA repair in B-cells, which deliberately fosters diversity in immunoglobulin production[18]. Yet demonstration of the DNA variants at the protein level of immunoglobulins, and their quantification is still limited[19]. Beyond antibodies, other instances of beneficial somatic mutations have been reported, though they remain relatively scarce and primarily confined to specific clinical contexts[20]. Still, as somatic variation is omnipresent and pervasive, unraveling its broader role in human physiology and adaptation represents a considerable challenge[6].

In addition to genetically encoded proteoforms, protein variants resulting from errors in transcription and/or translation ("phenotypic variants") represent a large fraction of the heterogeneous proteome. Phenotypic variants are orders of magnitude more prevalent than DNA mutations, ranging between $10^{-5}$ and $10^{-3}$-$10^{-4}$ per RNA and protein residues, respectively[21]. Originally mapped systematically in microbes[9,22], recent analyses were expanded to flies and human[23] revealing a codon usage effects and amino acid levels in cells

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as key determinants of translation infidelity. With ~$2\times10^9$ protein molecules in a human cell, and ~$10^{12}$ amino acids in them, an averaged translation error rate of $10^{-3}$ amounts to $10^9$ mistakenly incorporated amino acids, occurring in the cell's proteome[24]. A prime characteristic of phenotypic mutations is that they cast a significant burden on organisms. Proteins translated with errors tend to misfold, aggregate, engage in spurious interactions, saturate the protein quality control machinery[25], and cause diseases[26], e.g. neurodegeneration[27]. In parallel, an increasing number of reports indicate that some errors in translation can be advantageous at the cellular level [25,28,29]. In cancer, a recent summary provides evidence that translational errors contribute to the tumor's capacity to evolve[30]. On an evolutionary scale, phenotypic errors might open adaptive paths otherwise precluded by epistatic interactions[31], of purging of deleterious mutations[32], or expose hidden functionalities[33]. Whether deleterious or beneficial, the rates and nature of phenotypic mutations appear regulated[34] and the tendency to make or avoid them is postulated to be genetically encoded [9,35].

# Results

## Substitution search and validation pipeline

We developed an improved approach towards detection of amino acid substitutions within proteins, using mass spectrometric (MS) proteomic data. For detailed description see Supplementary Note 1 and Figure 1.

**Figure 1. Substitution strategy overview and outputs.** (A) Overview of the amino acid substitution detection pipeline utilized in this work. Raw proteomic data are split into identified and un-identified subsets, identified peptides are *in silico* mutagenized and searched for within the un-identified subset of the raw data. Substituted peptide candidates are validated by PepQuery2, localized by Ascore, verified for RNA-seq evidence in transcriptomics data and further filtered for the quantification purposes. (B) Substituted peptides identified by MaxQuant dependent-peptide search protocol[9] (orange), open-search based deTEL pipeline[36] (blue) and the method presented here (green) from the same dataset. (C) Relative abundances of substituted peptides and their reference variants as ratios of their precursor (MS1) intensities. Orange – substitutions without mRNA-seq support, blue – substitutions found in immunoglobulins (with or without mRNA-seq support), green – substitutions with mRNA-seq evidence. Antibody, organs, ribosome, mRNA and tRNA icons were created at BioRender.com by Dahan O.

We compared our pipeline with the previous method based on MaxQuant dependent peptide search as well as more recent MSFragger's open-search based protocol deTEL, and report increased quantity of unique identified peptides using strict validation criteria (detailed in Supplementary Note 1) [9,36]. For each detected substitution we assigned an abundance ratio



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defined as $\log_{10}$ ratio of precursor intensities of the substituted to reference peptide. We observed a wide range of substitution abundance ratios from $10^{-7}$ up to $10^{2}$ (Fig. 1c). In addition, we analyzed transcriptomic data provided with the original dataset[1] to determine potential genetic or non-genetic origin of the observed substitutions. Substitutions that have RNA-seq support tended to follow a multi-modal distribution, with highest level substitutions corresponding to 1:1, corresponding to heterozygocities, immunoglobulin substitutions centered around 1:100, and other substitutions centered around 1:10,000 (Fig. 1c).

## Amino acid substitutions are prevalent within the human proteome

We analyzed mass spectra from the healthy human tissues with available RNA-sequencing data and detected 13,910 amino-acid substitutions, of which 7,215 unique, complying with our filtering criteria. As a result of amino acid substitution-focus, we achieved at least 40% increase in number of detected peptides in comparison to previously reported methods, when applied to the same data (Supplementary Tables 1, 2 and 3)[9]. During the analysis we have discovered a systematic bias towards increased substitution detection rate near the tryptic peptide termini. This bias also existed in the variants supported by RNA sequencing representing genetically encoded substitutions, indicating that it does not represent false positive identifications within our data (Extended Data Fig. 3, left). Nevertheless, to avoid potential quantification bias associated with trypsin cleavage specificities [37], we only considered substitutions that were: (i) found on peptides derived from complete or near-complete (one missed cleavage maximum) trypsin digestion, (ii) within a >=3 amino acids away from N'- and C'-peptide termini and (iii) not originating/leading from/to lysine or arginine residues associated with trypsin activity. We refer to this dataset as the "filtered data", which we use only for analyses requiring MS1 abundance quantification (Extended Data Fig.3, right). This filtering retained 10,121 substitutions (out of which 5,257 are unique) (Supplementary Table 4). In cases where we only evaluated presence/absence of a substitution we employed the previously described "expanded set" of 13,910 substitutions (Supplementary Table 1).

Our collection of substitutions contains both genetically encoded single amino acid variants - SAAVs (due to either somatic mutations or heterozygotic allele expression) and phenotypic variants resulting from errors in transcription or translation. We utilized the RNA-seq data accompanied the proteomic dataset to dissect the origins of the observed SAAVs and determined if they are likely derived from DNA substitution, or phenotypically from

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transcription (or translation) errors. We assigned an RNA read coverage ratio for each MS1-derived substitution intensity. In analogy with MS substitution abundance ratio, it represents the $\log_{10}$ ratio of read counts encoding the observed SAAV to the counts of the reads encoding the reference amino acid. This ratio represents a direct measure of a genetic variability on mRNA level and an indirect estimation of it on the DNA level assuming the low frequency of transcription errors. In many cases, RNA coverage ratios suggest only small fraction of substituted transcript raising a challenge in distinguishing between biological and technical origin of substitution, such as error during transcriptomic library preparation. For each SAAV with RNA-seq evidence we filtered substitutions which could originate from transcriptomics pipeline error (see Methods section). This filtering yielded 5,358 RNA-Seq supported substitutions with high confidence of non-technical origin (Supplementary Table 1).

When the MS and RNA-seq derived substitution ratios are plotted against one another, several substitution groups emerge (Fig. 2). The first is centered around 0-0 coordinates, where both proteome MS and RNA-seq abundance ratios of the reference amino acid and the substituted one are about equal. Equality of the RNA read counts for the reference and substituted variant suggests heterozygosity of the individual sample donor, and about equal mRNA expression level of the two alleles. Interestingly the deviations from equal protein MS substitution abundance in this group may represent variable proteoform which may result from either from (i) differentially translated allelic variants or (ii) differential stability of the reference and alternative proteoforms.

**Figure 2. Quantification of substituted variants.** Comparison of substitution abundances provided by RNA-seq coverage and MS1 intensity ratios of substituted and reference peptides. Right – substitutions found on both protein and mRNA level and their corresponding abundances in comparison to the reference peptide and transcript. Grey background area - position of data present in expanded set but filtered out in the filtered set. Left – MS observed substitutions without evidence in RNA-seq data and sequencing coverage ≥ 1 supporting only the reference protein variant. Yellow – members of the immunoglobulin superfamily, blue – substitutions of all other (non-immunoglobulin) proteins. Violin plots on x/y axes represent distributions of immunoglobulin (yellow) and non-immunoglobulin (blue) coverage (x-axis) and abundance ratios (y-axis). (Top) immunoglobulin substitutions, split by whether a supporting genetic variant was detected (right) or not (left). Pie charts show substitution counts (below) and proportions (size) across CDR, framework (FWR), and constant regions (left to right). Colors indicate substitutions reachable by one (light blue), two (dark blue), or three (pink) codon mutations. Antibody icon was created at BioRender.com by Dahan O.

Next group in plot consists of the area spanning the values of -1.5 to -3 (substitution abundance ratios ranging from 1:10 to 1:1000) in both proteome, and RNA ratio. These substitutions

ACCELERATED ARTICLE PREVIEW

represent genetically encoded, low-abundance variants in the dataset. Although these variants could represent transcriptional errors, we suggest that many of these substitutions outline somatic DNA variation (since transcription infidelity is at much lower ratios ($10^{-4}$-$10^{-5}$ per transcribed nucleotide)[8]. Notably, ~50% (630 out 1,238) of the genetically encoded substitutions in this variant group were members of the immunoglobulin superfamily (Fig. 2, yellow). As immunoglobulins circulate systemically, we cannot disentangle secreted/delivered antibody variants from genuine local mistranslation events. We flag immunoglobulin detections in Supplementary Tables 1 and 4 and summarize their substitution trends in Fig. 2 (top).

The leftmost section of the plot represents all MS-observed substitutions which were not supported by RNA-seq data (Fig. 2, left). This group represents ~60% of the reported SAAVs. Variants in this region could be generated either by (i) RNA-seq coverage limitations, resulting in omission of supporting reads, (ii) unsampled somatic variants from different areas of the tissue, or (iii) ribosomal translation errors. Substitutions in this variant group had at least 1 RNA read confirming the reference amino acid but lacked RNA evidence for the alternative variant identified by MS (Fig. 2, left). A higher threshold on RNA coverage ≥10 reads mapped to the reference variant with zero reads mapped to the SAAV still yields 6,510 (76%) of total 8,552 substitutions without RNA-seq support for the SAAV. Although we cannot exclude a genetic origin of the detected substitutions, we assume that the majority are errors in translation or transcription as their frequency was evidenced to be orders of magnitude higher than the DNA mutation rate [8,9]. Indeed, the average $\log_{10}$ abundance ratio of these substitutions is -2.6 (~1/500, Fig. 1c) similar to the values reported in other analyses of translational fidelity [9,22,38].

Intriguingly, and in agreement with recent investigations of human proteomics data[38], 539 substitutions (2.3% of the 13,910 variants in our expanded set), showed an MS1 abundance ratio greater than zero, suggesting that these variants may exceed the abundance of their corresponding reference proteoforms (Supplementary Table 1). Even with stricter requirement for non-genetic substitution origin evidence by a minimum RNA sequencing coverage threshold as high as 100 RNA-seq reads, we identify 68 such non-immunoglobulin derived substitutions lacking genetic support (Supplementary Table 4). While the idea that non-coded proteins might abound in the proteome is intriguing, we acknowledge the limitations of MS1-based peptide quantification as well as potential confounding factors such as differential peptide ionization efficiency or chromatographic retention that may inflate apparent

ACCELERATED ARTICLE PREVIEW

abundance. Additionally, because most such detections arose from low-abundance proteins, quantification of abundance ratios may be biased by reduced MS1 signal-to-noise ratio.

Nevertheless, we experimentally assessed ionization efficiencies for seven reference/substituted synthetic peptide pairs selected from cases in which the substituted peptide exhibited higher MS1 intensity than its reference counterpart. In four out of seven pairs, the substituted variants showed 10-100× higher ionization efficiency than the corresponding reference peptides (Extended Data Fig. 4). However, the magnitude of these ionization efficiency differences was insufficient to fully account for the observed disparities in MS1 intensities. We therefore propose that a combination of factors (including differential ionization efficiency, noisy quantification at low signal levels, and chromatographic biases) may overestimate the abundance of substituted (non-genetic) forms by up to 100–1,000-fold relative to their reference variants. Accordingly, while the apparent abundance of substituted variants may be inflated by technical factors, their true levels are likely more modest. Importantly, they could still reach biologically meaningful abundances (on the order of single-digit percentages relative to the reference form) consistent with our ability to confidently detect and validate such variants even in low-abundance proteins.

## Expression ratios of genetically encoded protein variants recapitulate their human population frequency

We analyzed the abundance ratios of genetic SAAVs (heterozygous and somatic variants) considering their allele frequencies (AF) in the human population from the gnomAD 4.1 database[39] (Supplementary Table 5). To classify identified substituent as heterozygotic allelic variant, we required its $\log_{10}$ mRNA coverage ratio to be between –0.33 and 0.33 (maximum ~2-fold transcript abundance difference) and minimal sequencing coverage of 10 reads. In total 630 unique variants fit into these criteria, for 498 of them AF annotations are available in the gnomAD database. Most of these allelic variants expectedly show about equal protein abundances of reference and substituted variant (Fig. 2). Nonetheless, some variants showed markedly different protein abundance ratios than would be expected from their mRNA expression ratios. Remarkably, annotated rare variants in human population also tend to have lower abundance in comparison to the co-expressed reference variants corresponding to the more frequent allele variant. Despite potential confounding effect in differential ionization efficiency of SAAV peptides, we observed a positive correlation between the variant's human

ACCELERATED ARTICLE PREVIEW

population AF and its substitution abundance ratio (Pearson 0.24, p-value $8.1\times10^{-11}$, Fig 3a,d). This confirms and broadens suggested connection between the translation efficiency of transcripts with a missense mutation and its AF in the human population[40]. Alternatively, this result may also indicate a lower protein stability of variants that are rare in the human population.

Further, we analyzed AFs of the rest of the identified variants with RNA-seq support, that show a lower substituted-to-reference RNA ratio between -1.5 to -3, hypothesized to represent somatic mutants within the donor tissues. This subset is unlikely to represent heterozygosity because of the large deviation from 1:1 ratio of RNA expression. We identified 868 unique variants in 363 proteins. Notably, 432 variants originated from immunoglobulins (proteins associated with GO terms "immunoglobulin complex" or "antigen binding"), a protein family with extensive somatic mutations. Out of 868 unique substitutions, 363 (including immunoglobulins) were found in the human population as per gnomAD statistics. We observed a similar correlation between the allele frequencies and protein abundance ratio as in case of germline variants (Pearson 0.23, p-value $2.58\times10^{-10}$, Fig 3b,e). Our data suggest that the frequencies of *intra*-individual *de novo* somatic mutations recapitulate the *inter*-individual population AFs established by organismal selection. In other words, these data suggest that cells with somatic mutations that recapitulate more common SNPs in the human population are more likely to be retained or expand within the body, potentially due to fitness advantage or increased biological stability. Further, we analyzed similarly the "nongenetic" SAAV group that do not have support at the RNA level. There too we observed a positive correlation (Pearson 0.55 $p=1.88\times10^{-20}$) of the alternative to reference peptide MS1 abundance ratio and the human population AF of the corresponding SNP (Fig. 3 e,f). This indicates that non-genetic variant is more likely to occur/withstand degradation if it also happens to exist as a prevalent genetic variant.

**Figure 3. Relation of substitution abundance and its population frequency** (a) Subset of the abundance ratio plot (Fig 2) representing detected heterozygotic allele variants of proteins. RNA coverage ratio (-0.33,0.33) corresponds to ~twofold difference in the transcript abundance. Colored points represent substitutions also found in human population with the color corresponding to their allele frequency (AF in $\log_{10}$ scale, same color scheme as on 3b and 3c). Grey points represent substitutions not found in GnomAD database. (b) Subset of the abundance ratio plot representing somatic variants of the proteins. Colored points represent substitutions also found in human population with the color corresponding to their allele frequency. Grey points represent substitutions not found in GnomAD database. (c) Subset of the abundance ratio plot representing variants without RNA-seq evidence. Sequencing coverage for each substitution is ≥10 reads. Colored points represent substitutions found in

ACCELERATED ARTICLE PREVIEW

human population with the color corresponding to their minor allele frequency. (d) Correlation between population allele frequency and heterozygotic variant substitution abundance ratio. (e) Correlation between allele frequency in population and abundance ratio of somatic variants. (f) Correlation between population allele frequency and protein abundance ratio of variants without RNA-seq support. Pvalues in (d-f) were obtained by Pearson correlation significance test.

## Non-genetic variants likely consist of mistranslations and show ribosomal error patterns

Variants without RNA-sequencing support for the substituted sequence (6,510 substitutions with reference residue coverage ≥10, of which 3,561with the coverage ≥100 reads) represent the substantial part of the reported SAAVs. We assume that most of these substitutions consist of frequently occurring translation errors.[9,22,25].

Notably, the abundant substitutions observed in our and other analyses - I>V[41] , Y>F[42] and S>N[43] reflect high levels of mistranslation within specific substituted positions of proteins rather than a uniform trend across the proteome. Consequently, particular substitutions appear to be more prevalent in some protein contexts than in others, indicating a potential underlying mechanism involving either *cis* or *trans* regulation of translation errors. Our analysis revealed an unexpectedly high prevalence of C>A, C>S, H>D, M>D, and W>D substitutions across the dataset (Fig. 4b). These classes of substitutions were abundant among identifications from all three substitution detection pipelines tested here (Supplementary Tables 1, 2, and 3). Literature review suggests that these events can arise from technical chemical transformations: C>A/S has been associated with desulfurization during proteomic sample preparation [44]; H>D has been reported following sample oxidation [45]; and M>D and W>D has been observed under specific conditions such as harsh transition metal-catalyzed oxidation [46,47]. Thus, we consider this class of substitutions to potentially comprise a mixture of non-biological and biological events, as we consistently detect them in additional datasets that were not exposed to overt oxidative damage or desulfurization-promoting reagents (see further below). Accordingly, we flag these substitutions as cautionary in the main text (Fig. 4b) and supplementary data and treat them as non-biological unless supported by orthogonal evidence. All analyses of non-genetic variation from healthy tissue data presented here exclude these five substitution types (5,216 variants without RNA-seq support), unless stated otherwise.

The mistranslation spectrum was non-random and directional. The most frequent substitution classes included A>S, V>I/L, H>N, A>V, E>T, Y>F, A>T, and S>G. Directionality was

ACCELERATED ARTICLE PREVIEW

evident even among relatively conservative substitutions, including R>K vs. K>R, G>A vs. A>G, Y>F vs. F>Y, and T>S vs. S>T. Several observed classes have human-relevant precedent. The frequent A>S class is consistent with the conserved challenge of preventing serine-for-alanine mistranslation by alanyl-tRNA synthetase editing[48], whereas alanine-producing substitutions such as G>A, E>A, V>A, and T>A are supported by natural human error-prone tRNA variants [49,50]. Less frequent classes in our data also map onto known human mistranslation channels, including F>S from an endogenous human tRNA-Ser variant[51]. At the aggregate level, the most common origin residues were A, V, L, T, E, S, and G, whereas the most common destinations were N, I, S, V, T, G, D, and A. Origin totals combine opportunity in the detectable proteome with error propensity, while destination totals highlight recurrent mistranslation products (Supplementary Table 6).

To assess the functional relevance of non-genetic substitutions in the biological context we investigated four additional MS datasets of human cancer-derived cells cultured under tryptophan, tyrosine, leucine, or arginine depletion[52–54]. We identified both previously reported and novel substitutions under these conditions. We confirmed reported increase in W>F and Y>F under tryptophan and tyrosine depletion, respectively (Fig. 4c, left) [52]. We also corroborated the reported increase in R>K, R>C, R>H, and R>S substitutions under arginine depletion, extending the original analysis to a proteome-wide scale (Fig. 4c, top right). Notably, leucine depletion led more frequently to L>F and L>V (Fig. 4c, bottom right) in comparison to negative control. Additionally, leucine starvation also induced a prominent M>T substitution that was not detected in any non-starved control replicates (Fig. 4c, bottom right). Although M>T has been flagged as a potential sample preparation artifact associated with iodoacetamide (IAA) alkylation, its marked and specific enrichment in all replicates of leucine-starved cells is more consistent with a biological origin [55]. Finally, W>D was detected across all conditions, but a selective increase in W>D frequency was observed only under tryptophan/tyrosine starvation (Fig. 4c, left), suggesting that W>D may play a distinct physiological role under specific stress conditions. All detected substitutions in starvation datasets are reported in Supplementary Table 7 and complete 20×19 confusion matrices in Extended Data Fig. 5-7. For extended biochemical analysis of W>D and C>A substitutions, see Supplementary Note 2. Briefly, we used $^{18}O$ labeled water and MS analysis to determine that W>D occurs in the cell, not in the MS preparation, either due to mistranslation, tRNA misacylation or oxidative processes. Conversely, an analogous strategy, using $D_2O$ suggests that the C>A substitution

ACCELERATED ARTICLE PREVIEW

likely arise from TCEP-induced desulfurization during sample preparation rather than from mistranslation.

Next, we investigated the mutational distances between the codons of the reference and substituted residues in substitution cases lacking RNA-seq support. We examined whether the substitution patterns in non-genetic variants are distinct from those originating from mutations. We calculated the minimal nucleotide distance between the codons of the reference amino acid and the hypothetical codons for the observed substitution. We will refer to substitutions within mutational distance 1 as "near-cognates" and the rest as "non-cognates"[9]. We found that, among substitutions with RNA-seq support, the minimal nucleotide distance is 1 in most cases (5,001out of 5,130). By contrast, non-genetic variants frequently showed more distant substitutions - 2 nucleotides in 30% cases (1,578 out of 5,216) and occasionally 3 (119 out of 5,216, ~2%) nucleotides from the reference codon (Fig. 4d). Near-cognate non-genetic substitutions appear to be more abundant in comparison to non-cognate substitutions reflecting lower disruptive potential of the near-cognates on protein structure and stability (Fig. 4d)[9] . Consistent with these observations, the distribution of BLOSUM62 scores (evolutionary acceptance rate of the amino acid substitution between homologs) for non-genetic substitutions was shifted toward more negative values (*i.e.* lower observed evolutionary rate of substitutions), indicating a larger functional distance between the reference and substituted residues (Fig. 4e).

**Figure 4. Statistics of amino acid substitutions.** (a) Summary of counts of detected genetic substitution types (bubble size, relative to the total count of all substitutions) and their minimal mutational distance (bubble color), (b) summary of counts of detected non-genetic substitution types (bubble size) and their minimal mutational distance (bubble color), (c) significant changes in substitution counts between starved and control cells. Left – W/Y starvation, top right – L starvation, bottom right – R starvation, (d) substitution abundance distributions (with lines as 25$^{th}$, 50$^{th}$ and 75$^{th}$ percentile of the data) associated with different mutational distances in genetic (red) and non-genetic (blue) substitutions with sample sizes above each distribution , (e) substitution counts associated with distinct BLOSUM62 scores for genetic (red) and non genetic substitutions (blue)

## Abundant non-genetic substitutions provide means for functional protein diversification

We analyzed the potential functional impact of substitutions using AlphaMissense pathogenicity score (Supplementary Table 9)[56]. High pathogenicity score of the substitution

ACCELERATED ARTICLE PREVIEW

indicates a disturbance of a functionally important site with potential functional outcomes such as change of protein structure, activity or gain of a new function. Considering that substituted variants mostly represent the minority of the synthesized protein, we interpret their pathogenicity scores as a potential modulation of the main protein function, implemented by the reference proteoform. The higher pathogenicity score serves as a proxy for a potential "disruptiveness" of a given substitution. We performed a separate functional analysis on non-genetic and genetic substitutions in our dataset. To classify non-genetic substitution, we specified a minimum RNA sequencing coverage threshold as 10 reads per reference and 0 reads per substituted position. We observed a negative correlation between predicted pathogenicity and the substitution abundance ratio in both genetic and non-genetic variant groups, indicating that more disruptive substitutions tend to be present at lower levels in the proteome (Fig. 5a). For genetic variants, this relationship likely reflects the greater disruptive potential of low-abundance somatic substitutions relative to germline variants. The analogous trend among non-genetic variants is consistent with either (i) preferential synthesis of non-pathogenic variants or, more plausibly, (ii) enhanced degradation of more deleterious protein variants. Notably, correlation with predicted pathogenicity is weak and cannot fully explain observed abundance ratios. Moreover, non-genetic substitutions classified as pathogenic cannot impose a comparable fitness burden, as they are incorporated into a fraction of a protein's molecules rather than uniformly affecting all copies, as implicitly assumed by AlphaMissense. Such partial incorporation could expand functional diversity while avoiding the full fitness cost of a fixed genetic mutation.

**Figure 5. Functional characteristics of amino acid substitutions.** (a) AlphaMissense pathogenicity scores plotted against substitution abundances of genetic (left) and non-genetic (right) variants. (b) Distributions of alternative variant substitution abundances in proteasome-inhibited (left) and non-inhibited conditions (right). (c) Venn diagram of hemoglobin substitutions identified in three datasets. Red – substitutions found in healthy tissue dataset, orange – in purified hemoglobin isolate, blue – in in vitro differentiated hematopoietic precursor cell culture. (d) Positions with identified substitutions in human hemoglobin in healthy tissues dataset. Spheres indicate Cα position of the substituted residues. Color scheme corresponds to recurrence of a substitutions in each site among the tissues. Sites with associated changes in $O_2$ affinity highlighted

To examine whether alternative variants are subject to selective degradation, we leveraged a data in which osteosarcoma-derived cells were cultured in the presence or absence of the proteasome inhibitor MG132[57] . If alternative proteins were as stable as their reference forms, their relative abundance would be largely unchanged upon proteasome inhibition. Instead, we

ACCELERATED ARTICLE PREVIEW

observed a global increase in the abundance of alternative protein variants following proteasome inhibition, indicating preferential proteasome-dependent turnover (Fig. 5b, Extended Data Fig. 9 for replicate-separated abundance distributions). (further details in Supplementary Table 10).

Further focusing on functional consequences of the amino acid substitutions, we searched the expanded dataset of non-genetic substitutions for variations within the functional sites of proteins. We identified 85 unique positions in 22 proteins with substitutions within the annotated functional sites in UniProt database (Supplementary Table 11, only keywords "site", "binding site" and "active site" were considered, sample spectra Supplementary Fig. 1). We hypothesized that substitutions of residues proximal to annotated functional sites would also modulate protein activity. Substitutions in our dataset were significantly more clustered near annotated active sites than expected by chance ($p = 5\times10^{-5}$), whereas no significant clustering was detected around annotated binding sites ($p = 0.06$). This enrichment near active sites is consistent with the possibility that some mistranslations are functionally consequential, as substitutions close to catalytic residues are more likely to affect enzymatic activity. These results therefore support the idea that mistranslation can broaden the functional repertoire of the proteome by generating low-abundance variants with altered biochemical properties. Notably, many substituted functional-site residues involved C, H, M, and W - substitutions we previously flagged as "potentially non-biological." These substitution types were excluded from the presented proximity analysis. Nevertheless, because we cannot rule out a biological contribution, we report them alongside other functionally relevant substitutions in Supplementary Table 11, with uncertain origin indicated in the "ambiguous" column.

Next, we examined whether some of the non-genetic SAAVs in the expanded dataset, appear regularly across different samples. Recurrent appearance of the substitution across samples (*i.e.* across tissues and individuals) suggests its stability or perhaps potential functional significance. Further, such recurrence bolsters our confidence for its non-genetic origin, as the probability of absent sequencing evidence for a recurrent mutation in unrelated individuals is restrictively low. We gathered a set of 163 proteins and 392 unique substitutions which are repeatedly detected in 5 or more different samples (Supplementary Table 12, sample spectra in Supplementary Fig. 2-6).

ACCELERATED ARTICLE PREVIEW

As expected, a larger number of distinct substitutions was observed in highly abundant proteins. For example, we detected 227 and 44 unique variants in hemoglobin and GAPDH, respectively; of these, 46 hemoglobin substitutions and 10 GAPDH substitutions were observed recurrently in five or more individuals (Supplementary Table 12). To assess whether complex whole-tissue matrices may confound our identifications, we expressed these proteins in a purified form and subject them to a new MS measurement and substitution search. Specifically, we analyzed (i) 121 immunoprecipitated proteins from HEK293 cells generated as part of the BioPlex project and (ii) four highly purified proteins (hemoglobin, GAPDH, PRDX6, PARK7; >98% purity) produced recombinantly in, or isolated from, human cells [58]. Further, to test whether mistranslation is detectable on exogenous protein expressed in human, we analyzed a recombinant antibody produced and purified from HEK293 cells whose protein sequence shares no substantial similarity with any endogenous human protein (Supplementary Fig. 11,13,15). In comparison to healthy tissues samples, all these experimental datasets employed distinct sample preparation protocols and relatively fresh material, minimizing the likelihood of artifactual substitutions.

Across the 121 BioPlex immunoprecipitations, we detected 2,748 substitutions (median 21 per protein), including 54 variants that were also observed in our healthy-tissue analysis. In our analysis of purified proteins, we detected 60, 31, 27, and 27 substitutions in hemoglobin, GAPDH, PARK7, and PRDX6, respectively; of these, 37, 5, 3, and 4 were also detected in healthy tissues. All the overlaps were modest but statistically significant as per Fishers exact test (p-values $4.12\times10^{-21}$/$1.24\times10^{-9}$ for hemoglobin subunits, 0.02 for PARK7, $3.2\times10^{-5}$ GAPDH and $4.2\times10^{-5}$ PRDX6). Notably, both the BioPlex samples and our purified-protein preparations were freshly generated and MS sample preparation did not employ desulfurization-promoting protocols. Nevertheless, we again observed C>A, C>S, W>D, H>D, and M>D substitutions, supporting the possibility that at least a subset of these events may arise in cells. All substitutions identified in enriched samples from BioPlex project and purified proteins are reported in Supplementary Table 13 and 14 respectively and sample mirror spectra of co-identified peptides in healthy tissue samples and purified proteins on Supplementary Fig. 9 and 10.

Further, mass spectrometric analysis of the exogenous recombinant antibody prepared in HEK293 cells recovered 45 substitutions that could not be explained by the construct's coding

ACCELERATED ARTICLE PREVIEW

sequence, or human/bovine immunoglobulin contamination (IgBLAST, Supplementary Fig. 11,13). This orthogonal experiment confirms that mistranslation by the human ribosome is persistent and includes substitution with mutational codon distance >1 supported by high-quality MS/MS evidence (Extended Data Fig. 10, Supplementary Fig. 12,14 and Supplementary Table 14). In tissue proteomics, including the analyses reported here, hemoglobin is typically among the most abundant proteins due to tissue vascularization and residual blood content. As noted above, we detected 227 unique substitutions across the α- and β-subunits of hemoglobin (Fig. 5c,d). Some of these distinct substitutions were detected in multiple tissues yielding total 530 hemoglobin substitution events in the whole dataset (Supplementary Table 1). Hemoglobin function depends on an intricate allosteric network that enables the heterotetramer to transition between low- and high-affinity oxygen-binding conformations. Consequently, substitution in even a single subunit could yield hybrid tetramers with altered biological properties [59].

Among 227 unique substitutions, 27 matched exactly to characterized clinically relevant hemoglobin variants, including 20 associated with increased and 7 with decreased oxygen affinity (Supplementary Table 15). For these variants, the functional effects are supported by prior experimental characterization of the same amino-acid substitutions. A further 73 variants affected residues that are clinically relevant in hemoglobin but involved different amino-acid exchanges. We describe these as position-level functional candidates rather than validated functional variants with their precise functional effects remaining inferential (Supplementary Table 16).

We hypothesized that some of this diversity could arise genetically from somatic mutations accumulating across the limited number of hematopoietic lineages that ultimately generate mature red blood cells (RBC). Consistent with this possibility, among 530 non-unique hemoglobin substitutions detectable by MS, 82 (15%) were observed at low levels in RNA-seq data. However, the remaining MS-detected substitutions lacked corresponding RNA-seq support. To minimize potential confounding effects from genetic heterogeneity of the analyzed material and from RBC aging, we additionally analyzed a dataset of *in vitro*–differentiated erythroid precursors[60] . In this independent dataset, we detected 135 unique hemoglobin substitutions, 62 of which overlapped with those observed in healthy tissues (Figure 5d, Supplementary Table 17), representing a significant overlap (hypergeometric p-value = $4.06\times10^{-46}$) (Supplementary Fig. 7; spectra). These results independently confirm that

ACCELERATED ARTICLE PREVIEW

hemoglobin exhibits extensive substitution diversity *in vivo*, although the underlying mechanisms driving this diversification remain unclear.

In all these analyses we used the MS-supported substitutions lacking RNA-seq support as hypothesized mistranslations. We are aware that due to the RNA-seq coverage limitations we cannot completely exclude the cases of somatic mutations within the tissue samples. However, we assume that the averaged trends over different amino acid substitution patterns, their pathogenicity and allele frequencies would largely reflect the tendencies of protein mistranslation as it represents a major source of protein heterogeneity in cells[9,34]. Additionally, the existence of recurring substitutions in samples from non-related individuals allowed us to rationalize on distinct roles of non-genetic substitutions in human.

# Methods

## RNA-seq analysis

Raw RNA sequencing reads in fastq format were downloaded from Array Express (http://www.ebi.ac.uk/arrayexpress/experiments/E-MTAB-2836/) and their correspondence with the MS data was verified through Table EV1, tab "Sample Information" in Wang et *al*. publication[1] . Fastq files were aligned against ensembl CDS reference sequences (r112, https://ftp.ensembl.org/pub/release-112/fasta/homo_sapiens/cds/) using BWA[61] . Resulting files were converted, sorted and indexed using SAMtools[62] . Unmapped reads as well as supplementary, secondary and duplicated alignments were discarded. Observed codons at the genomic locations of each amino acid substitution were counted for each sample separately, and counts were summed per tissue. Alignments that contained N at the substitution position or an indel prior to it were removed.

We considered a substitution confirmed on a genetic level if observed number of reads with mutation could not arise from the DNA polymerase used in the sequencing library preparation. Specifically, we compute a one-sided binomial tail probability:

$$P(X \geq k|n,p) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

k = number of reads reporting the specific non-reference base that would produce the amino-acid substitution,

n = total RNA-seq read depth at that genomic position,

p = per-read probability of observing that specific base due to error.

ACCELERATED ARTICLE PREVIEW

We define p as:

$$p = \frac{\varepsilon}{3}$$

ε is the assumed per-base error rate (PCR/sequencing, 0.001 used in this manuscript based on reported values from [63] ), and division by 3 reflects that (under a simple symmetric model) an error at a base can convert to any of the other three nucleotides, but we are testing one specific alternative nucleotide.

Finally, we apply Benjamini–Hochberg correction across all substitutions tested to control FDR. The binomial test is performed at the RNA-seq level, the proteomics observation is not incorporated into the p-value calculation to avoid inflation of the dataset with the false positive genetic identifications.

## MSFragger search for the substituted candidates

Raw data for the analysis were obtained from Proteomic identification database (PRIDE ID PXD010154 for healthy tissue data, PXD027328 for degradomics dataset, PXD017276 for *in vitro* differentiated RBC precursors, PXD043612 for arginine starvation dataset, PXD044085 for leucine starvation dataset, PXD028921 for tryptophan/phenylalanine starvation dataset, ), BioPlex APMS datasets from MassIVE MSV000080679 repository and immunoglobulin profiling data from Zenodo database reference 10.5281/zenodo.14835053. and processed by FragPipe 23.1[64]. To separate search spaces for the reference and substituted peptide searches, a two-stage FDR-control protocol described previously was implemented. [65]. Raw data were searched using reference proteome database. Human protein sequences were taken from the Homo sapiens GRCh38 Ensembl release 112 peptide FASTA, together with the matched CDS/transcript FASTA, because our pipeline required codon-level mapping of candidate mistranslated residues. First search was performed using default parameters with MSBooster and IonQuant quantification activated with default settings[64,66]. Methionine oxidation was set as variable and cysteine carbamidomethylation as fixed modification. In case of degradomics datasets from Trulsson et *al.* additional variable modification of mass 114.04293 was set on lysine. "Write sub mzML" option in the Run tab of FragPipe was allowed in order to separate unidentified spectra and generate a workflow file for the following second search[65]. Automatic selection of best predictors of retention time and MS2 fragmentation spectra was chosen[67]. IonQuant was used for MS1 quantification with default parameters[68]. Match between runs was not activated.

ACCELERATED ARTICLE PREVIEW

Identified peptides from the first search were in silico mutagenized using custom python script and used as a reference database for the following search. The workflow file and new raw data in mzML format generated by a first search were used to perform a second search. The parameters were left default except: "Protein Digestion" on the MSFragger tab of FragPipe was changed to "nocleavage" to prevent further cleavage of the searched peptides, N-terminal methionine clipping was deactivated and minimum number of scans and isotopes for IonQuant quantification was decreased to 1. MSBooster was activated and retention time and MS2 fragmentation spectra predictors were set to the predictors selected in the first search (DIA-NN for the retention time prediction[69] and AlphaPept MS2 Generic for the fragmentation spectra prediction[70]). In case of $^{18}O$ labeled experiment, peptides were searched with allowed variable +2.0043 or +4.0085 Da modification on C-teminus and +4.0085 Da variable modification aspartate. In case of $D_2O$ labeled experiments, peptides were searched with allowed variable +1.0073 or +2.014 Da modification on C-/N-termini and +1.0073 Da variable modification on alanine.

## Validation of peptide candidates

First, spectral index was created from the MSFragger first search output - unidentified raw spectra in mzML format. Index was created as per PepQuery documentation. PepQuery was run in PSM validation mode in parallel for all identified PSMs on computational cluster. Input files for each parallel job consisted of tab separated values of the evaluated peptide sequence and its spectrum title reported in the psm.tsv file from the MSFragger first search. The validation search was set as follows: 20 ppm accuracy (-tol 20), 0-3 isotopic errors (-ti 0,1,2,3), fragment ion m/z tolerance of 0.05 Da (-itol 0.05) for Orbitrap and 0.6 for ion trap data. Enzyme cleavage was set to Trypsin (no P-rule, -e 2) and maximum two missed cleavages were allowed (-c 2). Search was not performed in the fast mode and scoring function was set to Hyperscore (-m 1). Validated peptide was required to have higher score than any of the competitors (-hc). Validation included competition with all possible post-translational modifications from UniMod database[71] as well as with single amino acid substitutions of the reference proteome peptides (-aa). The latter option caused that majority of the validated peptides were not reported (as they indeed represented a substituted reference peptides which are competed against in PepQuery validation) and were recorded in the ptm_detail.txt file together with all other potential PTM's and reference peptide variants. All reported variants for the tested peptides were extracted from the ptm_detail.txt. Peptide was considered validated if it had the highest score in comparison to the other candidates. PSM's with equally scoring candidates were

ACCELERATED ARTICLE PREVIEW

discarded. After the validation, all spectra were annotated graphically by PDV version 2.1.2 and b/y ion series were output in textual format. Only PSM's with at least two MS2 fragments per validated substituted position were considered for the further analysis. Additionally, substitutions found on the N-terminus of the peptide were not retained due to the low confidence of b1 ion identification[72]. Next, each spectrum of the identified alternative peptides was matched with its reference proteome counterpart using PepQuery without considering precursor mass (i.e. only on MS2 level). Spectra matching the reference peptides with higher score than alternative peptide matches were removed. Finally, all PSM's were processed by pyAscore algorithm with precision setting 0.01 Da for orbitrap and 0.6 Da for ion trap data. Only hits containing confident localization on the assumed substituted position were retained.

## Processing and filtering of the validated substituted peptides

For quantification, peptides were further filtered with the custom python script. Substitutions located on any position within 4 residues from the N' or C' termini were not considered due to potential trypsin cleavage bias[37]. Only correctly and singly mis-cleaved tryptic peptides were retained. Substitutions originating from or resulting in lysine or arginine were not considered due to quantification bias from the removal of existing or creation of a novel trypsin cleavage sites. For substitution abundance calculation, maximum observed intensity of the reference and alternative peptides of the same charge state was utilized. If reference and alternative peptide were found in several charged states, abundance ratio was calculated for every charged state separately and their median was taken to yield the final substitution ratio. This set of filtered substitutions was called "filtered data" and used for all analyses involving substitution abundance ratio.

## MaxQuant dependent peptide search and validation

Raw data were downloaded from Proteomic identification database (PRIDE ID PXD010154) and searched against Ensembl hg38 v112 proteome database using MaxQuant 2.4.13.0[73]. Dependent peptide search option was activated, match between runs deactivated. Carbamidomethylation of cysteine was set as fixed modification and methionine oxidations as a variable modification. AllPeptides.txt output table was processed by custom python scripts published by Mordret et *al*[9]. Raw files were converted to mzML format using ProteoWizard[74] and spectrum index for the following PepQuery validation was created according to the PepQuery documentation. Detected peptides from the dependent peptide search pipeline were searched again using PepQuery2 with the same settings as described in the methods section "Validation of peptide candidates" with exception of validation mode, tissue sample-specificity

ACCELERATED ARTICLE PREVIEW

and considering the alternative competitive peptides. Validation mode was set to simple peptide search (instead specific PSM validation). Peptides were queried along all the raw data disregarding tissue specificity. Identified peptides were not scrutinized against possible match with substituted variants of a different reference proteome peptide.

## AlphaMissense analysis

AlphaMissense predictions were obtained at https://zenodo.org/records/8208688 as AlphaMissense_aa_substitutions.tsv file. [49]Correlation between substitution abundance and their corresponding AlphaMissense pathogenicity was analyzed separately for genetic and non-genetic sets of substitutions.

## Functional site proximity analysis of non-genetic substitutions

To assess whether substitutions preferentially occur near functionally important residues, we performed a randomization test. For each observed substitution, we computed its linear distance (number of residues between the sites along the protein sequence) to the nearest annotated functional site and summarized these distances by the median across substitutions. We then compared the observed median distances to a null distribution generated from 20,000 simulated datasets in which substitution positions were randomized within the MS-observed regions of the corresponding protein sequences.

## GnomAD analysis

Each MS-observed amino acid was assigned a reference genome codon extracted from the CDS file associate with v112 ensembl genome release. Genomic coordinates for each codon corresponding to the MS-observed substitution were extracted with ensembldb utility in R[75]. The table containing genomic coordinates for the MS-observed substitutions was joined with GnomAD v 4.1 exome data downloaded at https://gnomad.broadinstitute.org/ containing collection of SNPs, their genomic location and their corresponding population allele frequencies[39]. This joined table was filtered only for such SNPs which would result in MS-observed missense substitution using custom python script. The substitution abundances were evaluated considering their hypothetical allele frequencies within the human population. Three groups of substitutions were evaluated – heterozygotic alleles, somatic variants and variants without RNA support. All variants with sequencing support were filtered towards elimination of substitutions derived by Illumina error. Heterozygotic variants were required that their coding transcripts coverage ratio lie between log10 values between –0.3 and 0.3 corresponding to two-fold difference in mRNA expression and additionally that sequencing coverage of was

ACCELERATED ARTICLE PREVIEW

at least 10. Somatic variants were required to lie between coverage ratio -1.5 and -3.5 corresponding to 3% and 0.03% of substitution-supporting reads within the number of reads covering that position.

## Immunoglobulin analysis

The sequences of proteins with MS-observed substitutions and which associated with GO terms "immunoglobulin complex" and "antigen binding" (GO codes) were analyzed by the IgBlast tool [76]. Blastp program was used with KABAT as variable domain delineation system [77]. The tabular output was parsed by custom python script to extract the variable domain regions from the alignment (CDR's and FR's). Substitutions with and without genetic support were analyzed for their location within or outside the immunoglobulin annotated region.

## Calculation of amino acid substitution fractions

Amino acid counts from the unique peptides yielded from reference proteome search (first search of the pipeline) for all the analyzed tissues were aggregated together. Next all unique amino acid substitutions originating from each kind of amino acid throughout all the samples were also counted. Substitutions fraction for a given pair of origin and destination amino acid was calculated as a fraction of observed substituted origin amino acids to the total number of observed amino acids of this kind.

## Synthetic peptide analysis

All synthetic peptides were ordered from GenScript (>=98% purity). Their sequences are listed in the Supplementary Table 18. The peptides were dissolved to 1nmol/µL separately and then mixed together to a stock of 10pmol/µL per peptide. The sample was then further diluted to 50 fmol/ul, and 9 ul were injected into the instrument. The resulting peptides were analyzed using nanoflow liquid chromatography (nanoAcquity) coupled to high-resolution, high mass accuracy mass spectrometry (Fusion Lumos). Data was searched against the list of peptides using the Byonic search engine and quantified using Skyline.

## Proteomic sample preparation

Pure proteins (GAPDH (MedChemExpress, HY-P78776), PRDX6 (CusaBio, CSB-MP018659HU), PARK7 (HUABio, HUA-HA211158) and Hemoglobin (MP-Bio, MPB-0855914)) were dissolved from lyophilized powder with 5% SDS in 50 mM Tris-HCl. 30ug was used for digestion

ACCELERATED ARTICLE PREVIEW

Samples were prepared according to the previously published protocol optimized for increased digestion efficiency [78] . Samples were reduced with 5 mM dithiothreitol and alkylated with 10 mM iodoacetamide in the dark. Phosphoric acid was added to the lysates to a final concentration of 1.2%, followed by the addition of 90% methanol in 5 mM ammonium bicarbonate. Each sample was then loaded onto an S-Trap 96-well plate (Protifi, USA). Samples were then digested with trypsin (1:50 trypsin/protein) O/N h at 37 °C. The digested peptides were eluted using 50 mM ammonium bicarbonate; trypsin was added to this fraction and incubated 4hr at 37 °C. Two more elutions were made using 0.2% formic acid and 0.2% formic acid in 50% acetonitrile. The three elutions were pooled and vacuum centrifuged to dry. Samples were kept at −20 °C until analysis. In case of $^{18}O$ and $D_2O$ labeled experiments, all reagents were freshly dissolved in labeled water.

## High pH Reverse Phase fractionation

UPLC/MS grade solvents were used for all chromatographic steps. 100ug of the whole cell lysates and 27ug of the purified protein digests were fractionated using High Performance Liquid Chromatography (Acquity H Class Bio, Waters, Milford, MA, USA). Mobile phase was: A) 20 mM ammonium formate pH 10.0, B) acetonitrile. Peptides were separated on an XBridge C18 column (3x100mm, Waters) using the following gradient: 3% B for 2 min, linear gradient to 40% B in 50 min, 5 min, to 95% B, maintained at 95% B for 5 min and then back to initial conditions. A total of 90 Fractions were collected (one fraction per 30s) from between 5-50 minutes and combined to a total of 15 fractions. Fractions were then analyzed by MS.

## Mass spectrometry

Samples undergoing DDA acquisition were fractions from High pH RP or the synthetic peptides. UPLC/MS grade solvents were used for all chromatographic steps. Each sample was loaded using split-less nano-Ultra Performance Liquid Chromatography (15 kpsi M-Class nanoAcquity; Waters, Milford, MA, USA). The mobile phase was: A) $H_2O$ + 0.1% formic acid and B) acetonitrile + 0.1% formic acid. Desalting of the samples was performed online using a reversed-phase Symmetry C18 trapping column (180 µm internal diameter, 20 mm length, 5 µm particle size; Waters). The peptides were then separated using a T3 HSS nano-column (75 µm internal diameter, 250 mm length, 1.8 µm particle size; Waters) at 0.35 µL/min. Peptides were eluted from the column into the mass spectrometer using the following gradient: 4% to 30%B in 55 min, 30% to 90%B in 10 min, maintained at 90% for 7 min and then back to initial conditions. Samples were run in a random order.

ACCELERATED ARTICLE PREVIEW

The M-Class nanoUPLC was coupled online through a nanoESI emitter (20 µm tip; Fossil Ion Tech, Madrid, Spain) to an Exploris 480 (for 2D fractions) or tribrid Orbitrap Fusion Lumos (for synthetic peptides) mass spectrometer (Thermo Scientific) using a PicoView nanospray apparatus (New Objective).

Data was acquired in data-dependent acquisition (DDA) mode, using a Top-Speed (2sec) method. MS1 resolution was set to 120,000 (at 200m/z), mass range of 375-1500m/z, AGC set to 200% and maximum injection time was set to Auto. Minimum intensity to 5e4, charges limited to 2-6 and dynamic exclusion to 40s. MS2 resolution was set to 15,000, quadrupole isolation to 1.6 m/z, AGC to 75%, maximum injection time to Auto, and HCD to 30 NCE.

### Statistics and Reproducibility

All experiments were performed in one replicate as the purpose of the analysis was to demonstrate the existence of substituted species and eventually to validate specific substitution types also identified in public data. No quantitative claims were raised.

### Graphics

Selected icons were created by O.Dahan using BioRender.com. Organs (Fig 1A) - *https://BioRender.com/5l6nzom; ribosome (Fig 1C)* - *https://BioRender.com/6ozp6rp; mRNA (Fig 1A)* - *https://BioRender.com/wzjcs3i*; mRNA (Fig 1C) - *https://BioRender.com/88yiadq*; antibody (Fig 1C and Fig 2) - *https://BioRender.com/ mhmqdpo*

# Conclusion

In conclusion, our data reveal that unity and multiplicity coexist within the proteome: the genome's unique signature is represented not just by a single encoded form, but rather by a spectrum of functionally relevant proteoforms that blur the line between individual and population. Our study, therefore, provides a unique perspective on an additional layer of genetic design complexity, highlighting the existence of a functional, recurrent, and evolvable protein quasi-species within the human proteome.

# Figure legends

ACCELERATED ARTICLE PREVIEW

**Figure 1. Substitution strategy overview and outputs.** (A) Overview of the amino acid substitution detection pipeline utilized in this work. Raw proteomic data are split into identified and un-identified subsets, identified peptides are *in silico* mutagenized and searched for within the un-identified subset of the raw data. Substituted peptide candidates are validated by PepQuery2, localized by Ascore, verified for RNA-seq evidence in transcriptomics data and further filtered for the quantification purposes. (B) Substituted peptides identified by MaxQuant dependent-peptide search protocol[9] (orange), open-search based deTEL pipeline[36] (blue) and the method presented here (green) from the same dataset. (C) Relative abundances of substituted peptides and their reference variants as ratios of their precursor (MS1) intensities. Orange – substitutions without mRNA-seq support, blue – substitutions found in immunoglobulins (with or without mRNA-seq support), green – substitutions with mRNA-seq evidence. Antibody, organs, ribosome, mRNA and tRNA icons were created at BioRender.com by Dahan O.

**Figure 2. Quantification of substituted variants.** Comparison of substitution abundances provided by RNA-seq coverage and MS1 intensity ratios of substituted and reference peptides. Right – substitutions found on both protein and mRNA level and their corresponding abundances in comparison to the reference peptide and transcript. Grey background area - position of data present in expanded set but filtered out in the filtered set. Left – MS observed substitutions without evidence in RNA-seq data and sequencing coverage ≥ 1 supporting only the reference protein variant. Yellow – members of the immunoglobulin superfamily, blue – substitutions of all other (non-immunoglobulin) proteins. Violin plots on x/y axes represent distributions of immunoglobulin (yellow) and non-immunoglobulin (blue) coverage (x-axis) and abundance ratios (y-axis). (Top) immunoglobulin substitutions, split by whether a supporting genetic variant was detected (right) or not (left). Pie charts show substitution counts (below) and proportions (size) across CDR, framework (FWR), and constant regions (left to right). Colors indicate substitutions reachable by one (light blue), two (dark blue), or three (pink) codon mutations. Antibody icon was created at BioRender.com by Dahan O.

**Figure 3. Relation of substitution abundance and its population frequency** (a) Subset of the abundance ratio plot (Fig 2) representing detected heterozygotic allele variants of proteins. RNA coverage ratio (-0.33,0.33) corresponds to ~twofold difference in the transcript abundance. Colored points represent substitutions also found in human population with the color corresponding to their allele frequency (AF in $\log_{10}$ scale, same color scheme as on 3b and 3c). Grey points represent substitutions not found in GnomAD database. (b) Subset of the abundance ratio plot representing somatic variants of the proteins. Colored points represent substitutions also found in human population with the color corresponding to their allele frequency. Grey points represent substitutions not found in GnomAD database. (c) Subset of the abundance ratio plot representing variants without RNA-seq evidence. Sequencing coverage for each substitution is ≥10 reads. Colored points represent substitutions found in human population with the color corresponding to their minor allele frequency. (d) Correlation between population allele frequency and heterozygotic variant substitution abundance ratio. (e) Correlation between allele frequency in population and abundance ratio of somatic variants. (f) Correlation between population allele frequency and protein abundance ratio of variants without RNA-seq support. Pvalues in (d-f) were obtained by Pearson correlation significance test.

**Figure 4. Statistics of amino acid substitutions.** (a) Summary of counts of detected genetic substitution types (bubble size, relative to the total count of all substitutions) and their minimal mutational distance (bubble color), (b) summary of counts of detected non-genetic substitution types (bubble size) and their minimal mutational distance (bubble color), (c) significant

ACCELERATED ARTICLE PREVIEW

changes in substitution counts between starved and control cells. Left – W/Y starvation, top right – L starvation, bottom right – R starvation, (d) substitution abundance distributions (with lines as 25$^{th}$, 50$^{th}$ and 75$^{th}$ percentile of the data) associated with different mutational distances in genetic (red) and non-genetic (blue) substitutions with sample sizes above each distribution , (e) substitution counts associated with distinct BLOSUM62 scores for genetic (red) and non genetic substitutions (blue)

**Figure 5. Functional characteristics of amino acid substitutions.** (a) AlphaMissense pathogenicity scores plotted against substitution abundances of genetic (left) and non-genetic (right) variants. (b) Distributions of alternative variant substitution abundances in proteasome-inhibited (left) and non-inhibited conditions (right). (c) Venn diagram of hemoglobin substitutions identified in three datasets. Red – substitutions found in healthy tissue dataset, orange – in purified hemoglobin isolate, blue – in in vitro differentiated hematopoietic precursor cell culture. (d) Positions with identified substitutions in human hemoglobin in healthy tissues dataset. Spheres indicate Cα position of the substituted residues. Color scheme corresponds to recurrence of a substitutions in each site among the tissues. Sites with associated changes in $O_2$ affinity highlighted

# Data availability

Raw data was submitted to PRIDE repository under ID PXD073135. Publicly available raw mass spectrometric datasets from https://www.proteomexchange.org/
PXD010154 healthy tissue data, PXD027328 degradomics dataset, PXD017276 in vitro differentiated RBC precursors, PXD043612 arginine starvation dataset, PXD044085 leucine starvation dataset, PXD028921 tryptophan/phenylalanine starvation dataset, PXD073135.

Publicly available raw mass spectrometric datasets from
https://massive.ucsd.edu/ProteoSAFe/static/massive.jsp
MSV000080679 - BioPlex APMS

# Code availability

The substitution search pipeline described here is based on running several third-party freely accessible tools (FragPipe first and second search, PepQuery index building and validation, labile PTM filtering, localization scoring and PDV annotation). We provide a detailed general guidelines on running these software packages, data analysis and evaluation in the methods section. Additionally, we provide example scripts used for complete substitution search and validation on https://github.com/treko90/tarsus.

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ACCELERATED ARTICLE PREVIEW

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# Competing interests

Authors declare no competing interests

# Author contributions

YP and VT designed and supervised the study; VT, TL and OA performed data analysis, DM and YL did the protein and peptide mass spectrometry, VT and YP wrote a draft, OA/DD/DM/OD consulted the study and critically reviewed the text. All authors approved the manuscript.

Supplementary Information is available for this paper.

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# Extended data figures legends

**Extended data Figure 1. Analyses of peptide retention time and MS2 spectra predictions.** Comparisons of predicted and experimental retention times (left column) and fragmentation spectra separated by distinct isotopic offsets of MS1 precursor (middle column) and precursor charge (right column). Comparisons presented for reference proteome peptides (a,b,c); alternative variants without genetic support (d,e,f) and alternative variants with genetic support (g,h,l). The inner boxes of the violin plots show the median (50th percentile; centre line), first and third quartiles (25th and 75th percentiles; lower and upper box bounds), and whiskers extending to the most extreme observations within of the quartiles. Points beyond the whiskers represent outliers. “n” under the violins stand for a number of distinct peptide-spectrum matches associated with plotted values of predicted spectral similarity.

**Extended data Figure 2. Examples of predicted and observed substitution supporting spectra.** Mirror plots representing predicted (bottom) and observed (top) fragmentation spectra of alternatively substituted precursors. Sample demonstrates spectra of substitutions within mutational distance >1. (a,b,c) Spectra well aligned with the prediction; (d,e,f) spectra with lower similariy with prediction

**Extended Data Fig. 3: Evaluation of trypsin cleavage bias in peptide termini.** Distribution of positions within the tryptic peptide length where substitution was detected. Right – substitutions within the “extended set”, left – substitutions in “filtered set”. Orange – substitutions mapped to RNA sequencing data, blue – substitutions without RNA sequencing support

**Extended Data Fig. 4: Detectability evaluation of synthetic peptides with increased abundance in healthy tissue data.** LC–MS analysis of seven synthetic peptide pairs selected from cases where the alternative (non-genetic) form appeared unusually abundant relative to the reference in our dataset. Substituted peptides are shown in red and reference peptides in grey. log10(A/R)syn denotes the detectability ratio of alternative/reference peptides measured from an equimolar synthetic mixture. log10(A/R)tis denotes the log10 substitution-abundance ratio of alternative/reference peptides measured in the healthy-tissue dataset.

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**Extended Data Fig. 5: Analysis of arginine starvation dataset.** Amino acid substitution matrices with counts (numbers in cells) and substitution fraction (color) of the specific substitutions in control (a,b,c) and arginine depleted (d,e,f) breast cancer MB-231cells. Substitution fraction is defined as ratio of substituted origin amino acids to its total observed count in the MS data.

**Extended Data Fig. 6: Analysis of leucine starvation dataset.** Amino acid substitution matrices with counts (numbers in cells) and substitution fraction (color) of the specific substitutions in control (a,b) and leucine depleted (c,d) breast cancer MB-231cells. Substitution fraction is defined as ratio of substituted origin amino acids to its total observed count in the MS data.

**Extended Data Fig. 7: Analysis of tryptophan/tyrosine starvation dataset.** Amino acid substitution matrices with counts (numbers in cells) and substitution fraction (color) of the specific substitutions in control (a,b), tryptophan depleted (c,d) and tyrosine depleted (e,f) melanoma MD55A3 cells. Substitution fraction is defined as ratio of substituted origin amino acids to its total observed count in the MS data.

**Extended Data Fig. 8**: **Analysis of W>D substitution-bearing peptides in three independent proteomic datasets.** (a) Venn diagram showing the number of W>D identifications in each dataset, with overlaps indicating peptides detected in common. (b–f) Annotated MS/MS spectra of the five W>D peptides identified across all three datasets ($^{18}$O-labeled HeLa cells, healthy tissues, and W/Y-starved human cell lines). Some W>D peptides were detected in multiple healthy tissues and starvation conditions (top left corner of each spectrum). Similarity of the experimental and predicted spectra in the top right corner of each spectrum. Fragments supporting the substitution are highlighted in red, the site-determining fragment is marked with a black star, and +2 Da / +4 Da suffixes on y-ion labels indicate the presence of one or two $^{18}$O atoms at the precursor C-terminus.

**Extended Data Fig. 9: Analysis of proteasome inhibition dataset.** (Left) Violin plots showing the distributions of substitution abundances in untreated U2OS cells (blue) and MG-132–treated U2OS cells (red), shown separately for each biological replicate. The middle line within the violin shows the median. Number of observations in each replicate are indicated as N above each respective violin.

**Extended Data Fig. 10**: **Substitutions detected in recombinant antibody**. Substitutions detected in heavy (top) and light (bottom) chains of affinity purified human recombinant immunoglobulin protein R1. Amino acid positions covered by MS highlighted by the green panel above the x-axis. Each bar represents the count of different substitutions detected in respective positions of the protein. CDR areas highlighted in orange, framework regions in purple/green. Inset – proportions of the mutational distances among detected substitutions of each chain.

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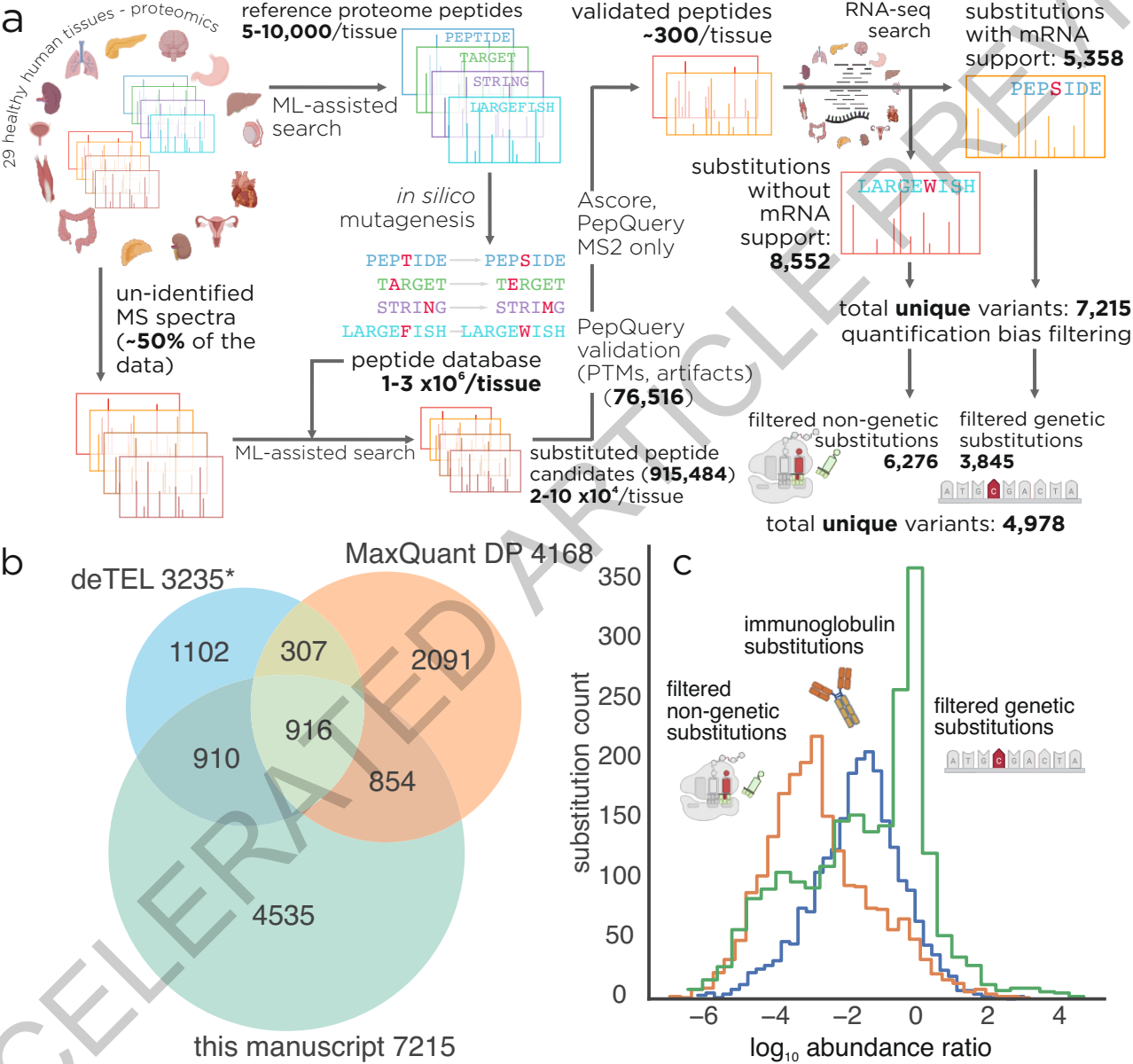


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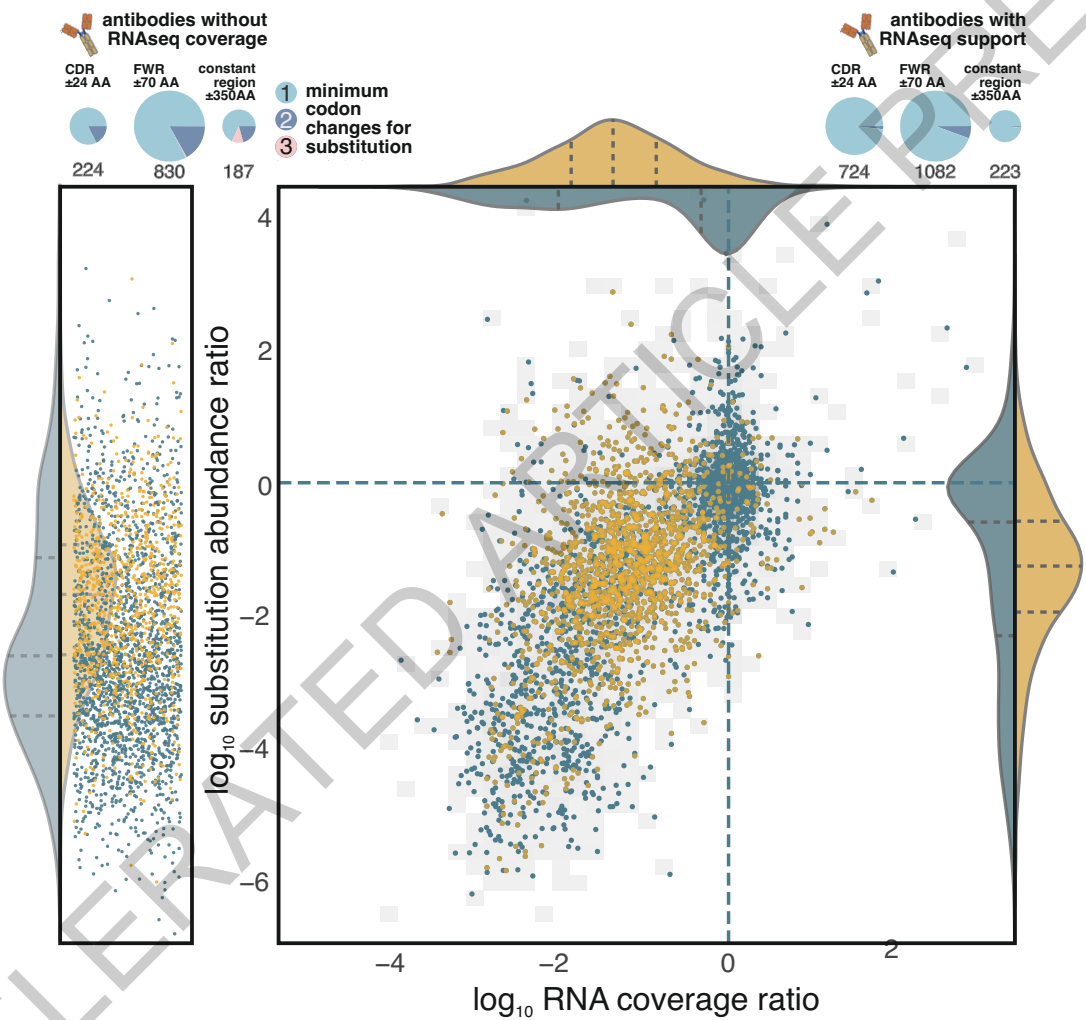
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FWR ±70 AA
constant region ±350AA
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antibodies with RNAseq support
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FWR ±70 AA
constant region ±350AA
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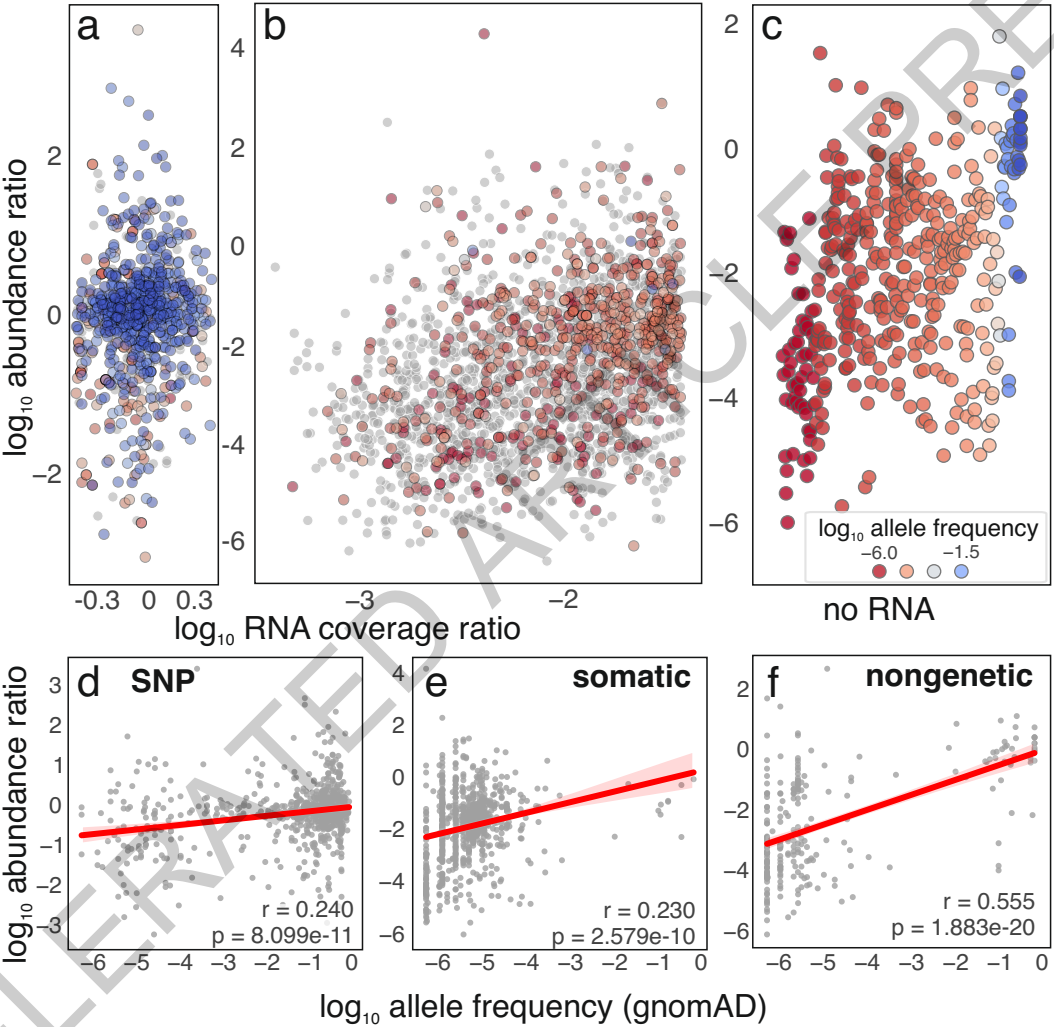


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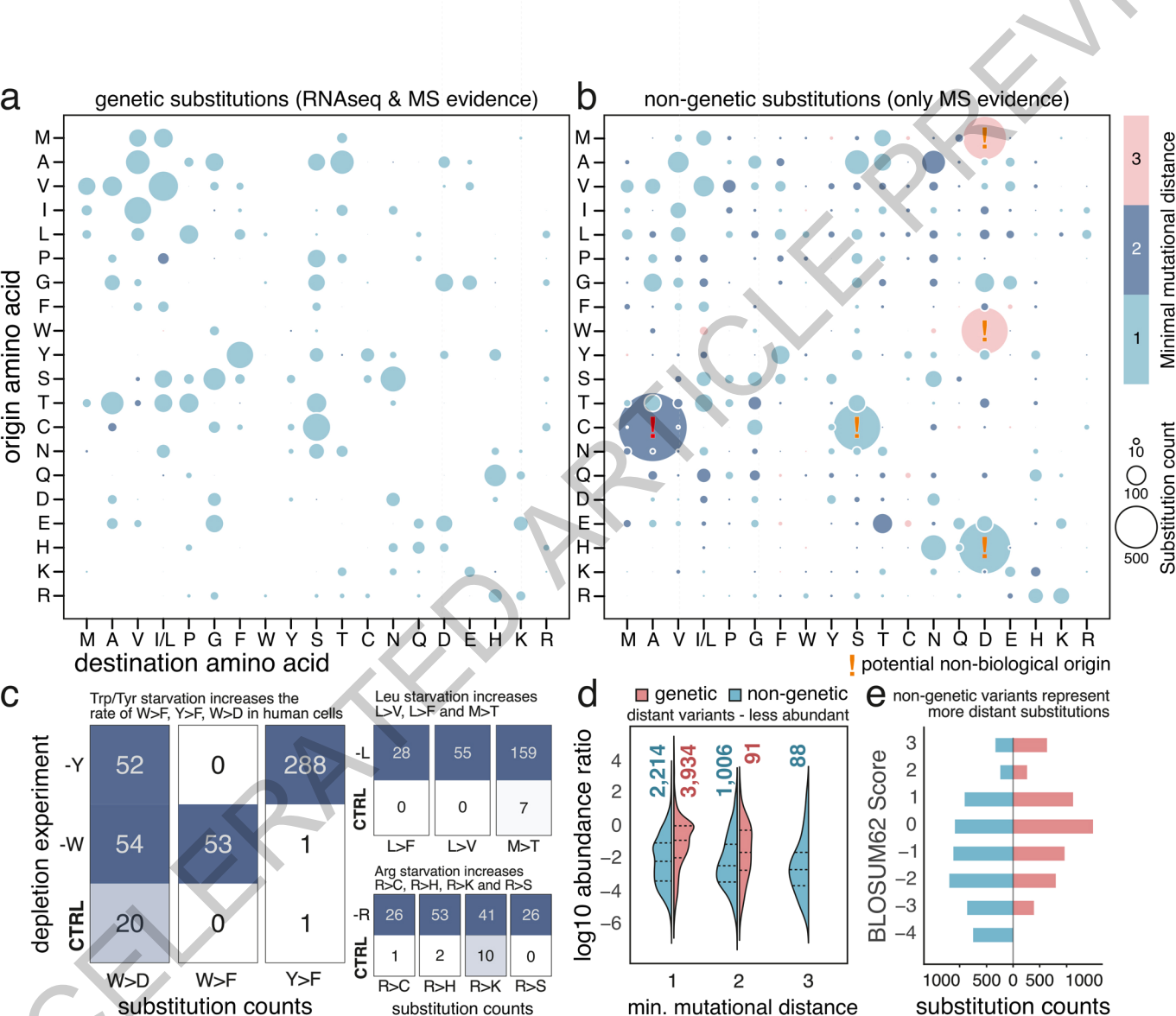


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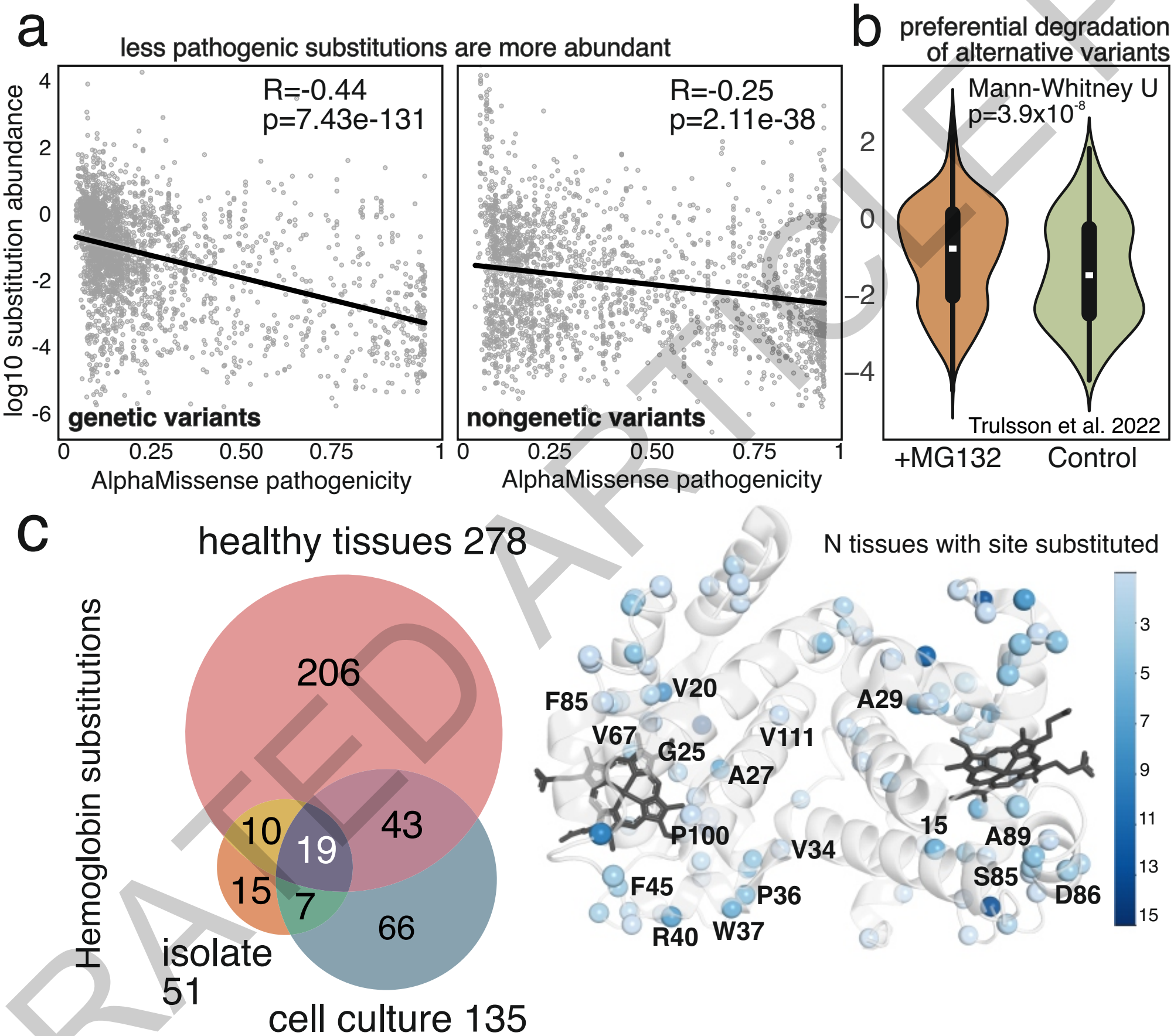
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log10 substitution abundance
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p=7.43e-131
genetic variants
AlphaMissense pathogenicity
R=-0.25
p=2.11e-38
nongenetic variants
AlphaMissense pathogenicity
b
preferential degradation
of alternative variants
Mann-Whitney U
p=3.9x10-8
Trulsson et al. 2022
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Control
c
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Hemoglobin substitutions
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isolate
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cell culture 135
N tissues with site substituted
F85
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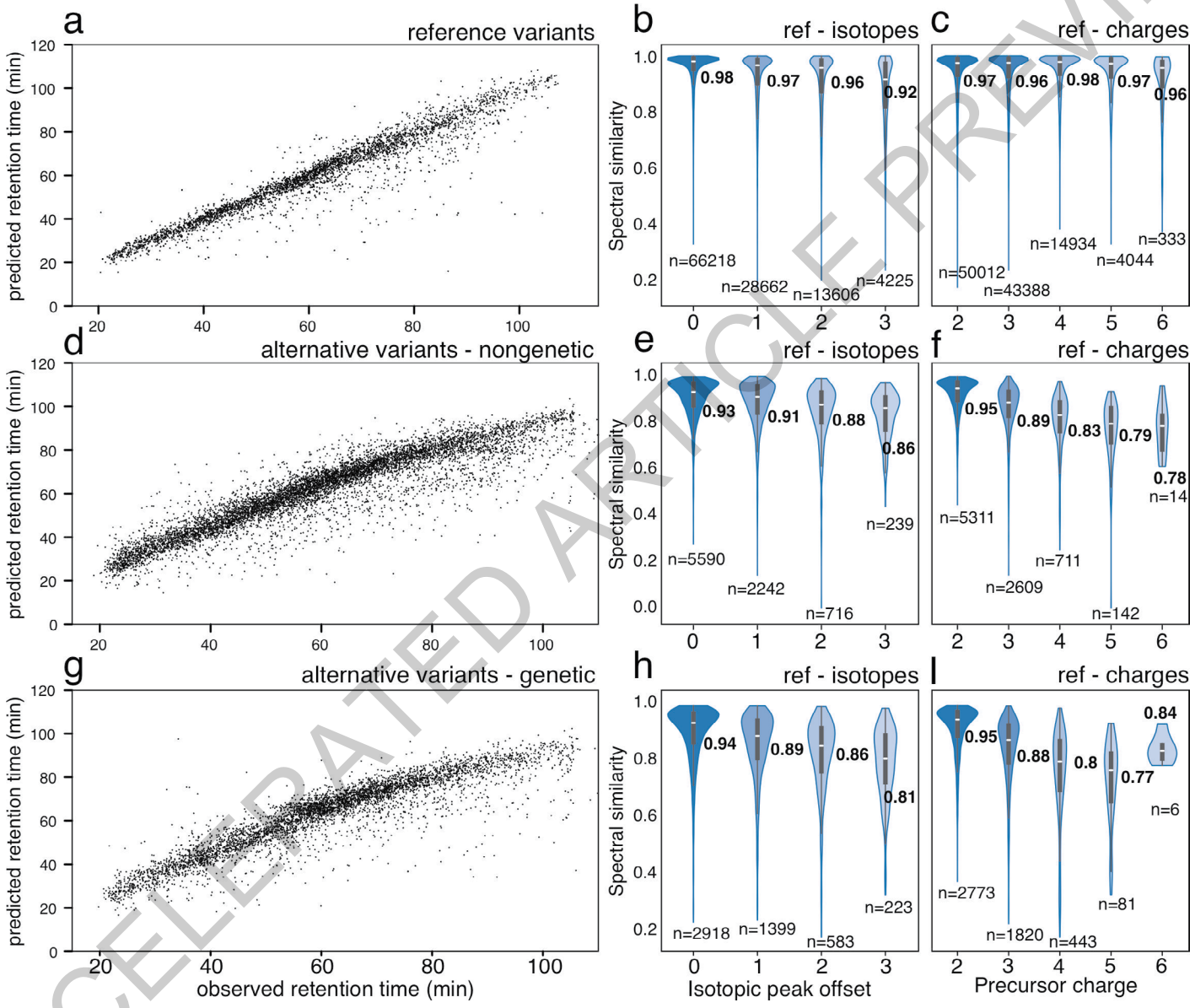


**Extended Data Fig. 1**

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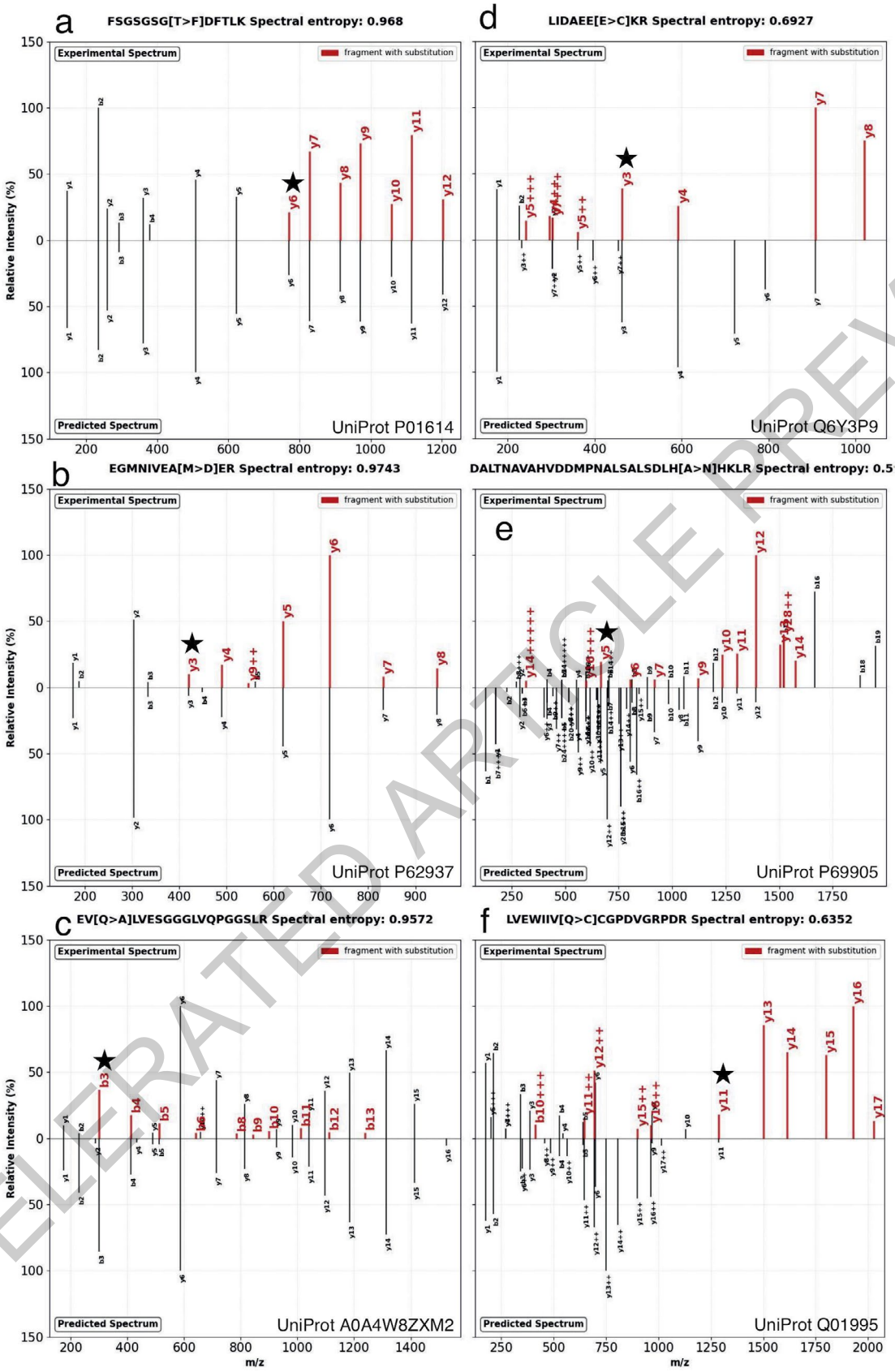


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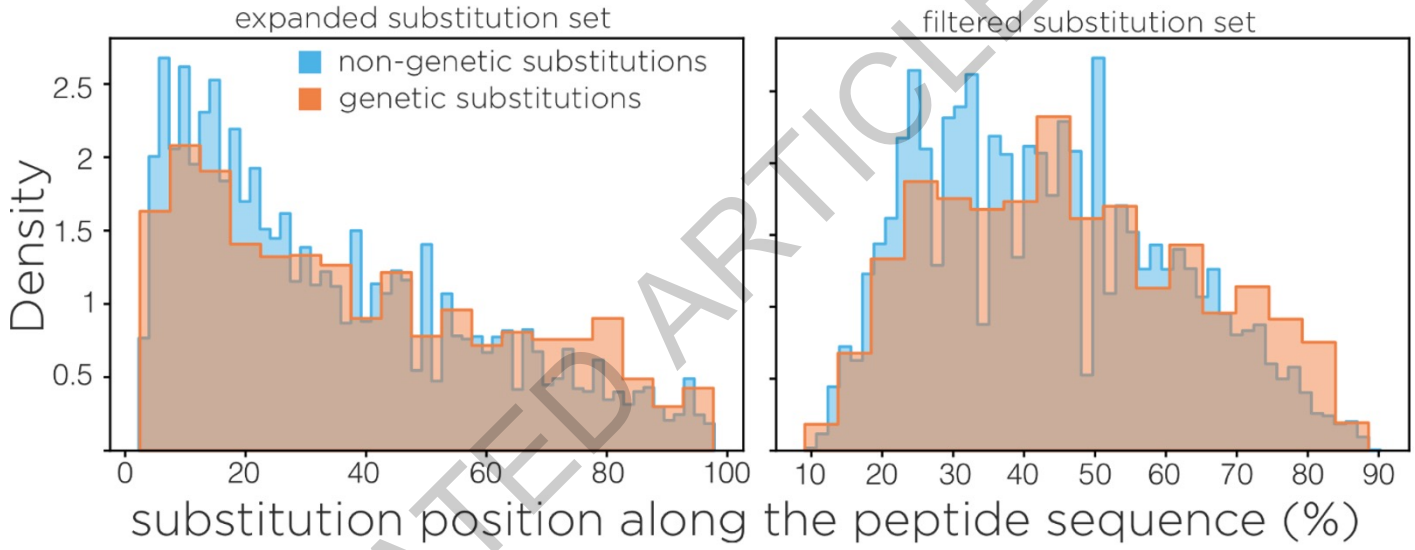


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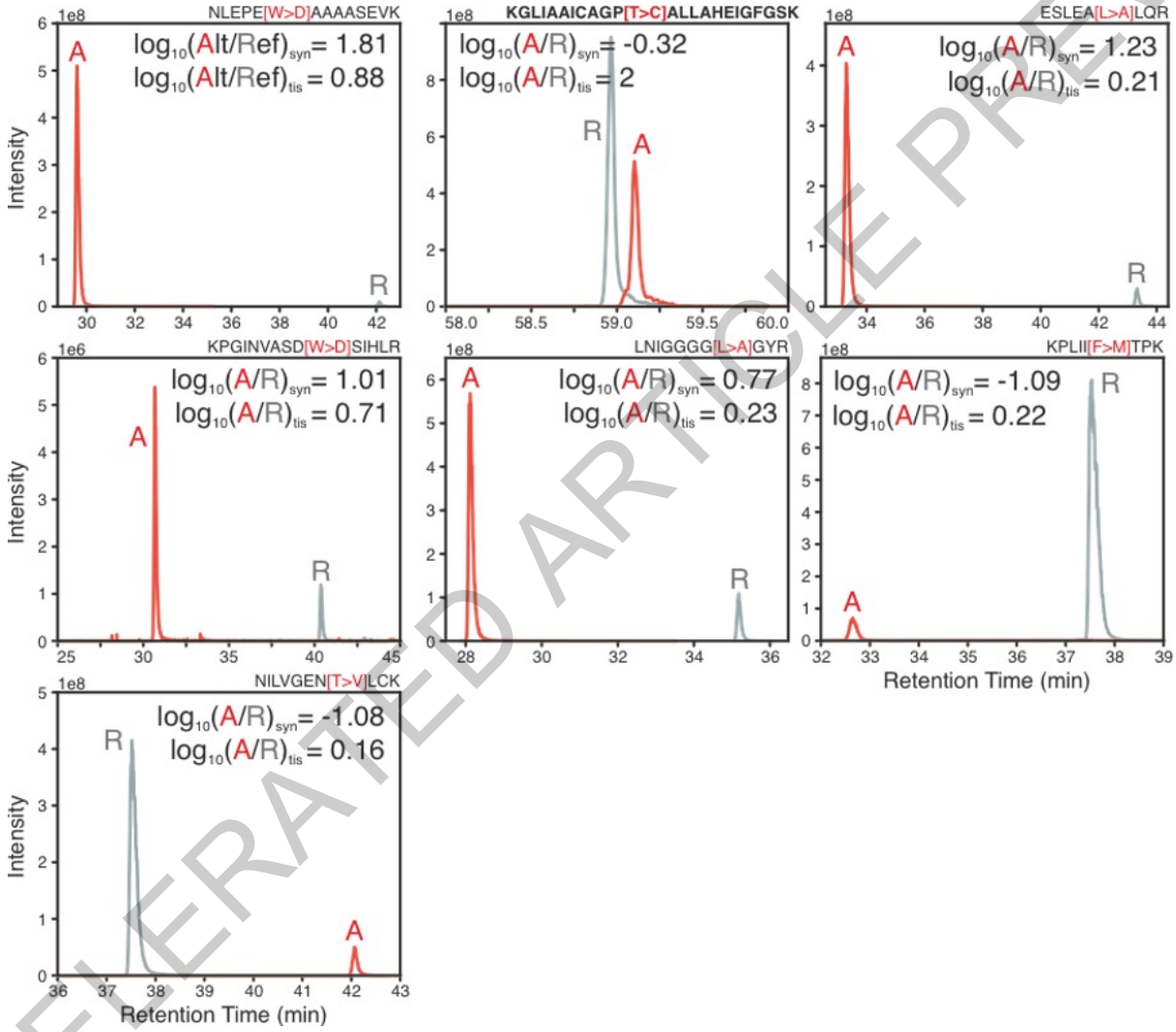


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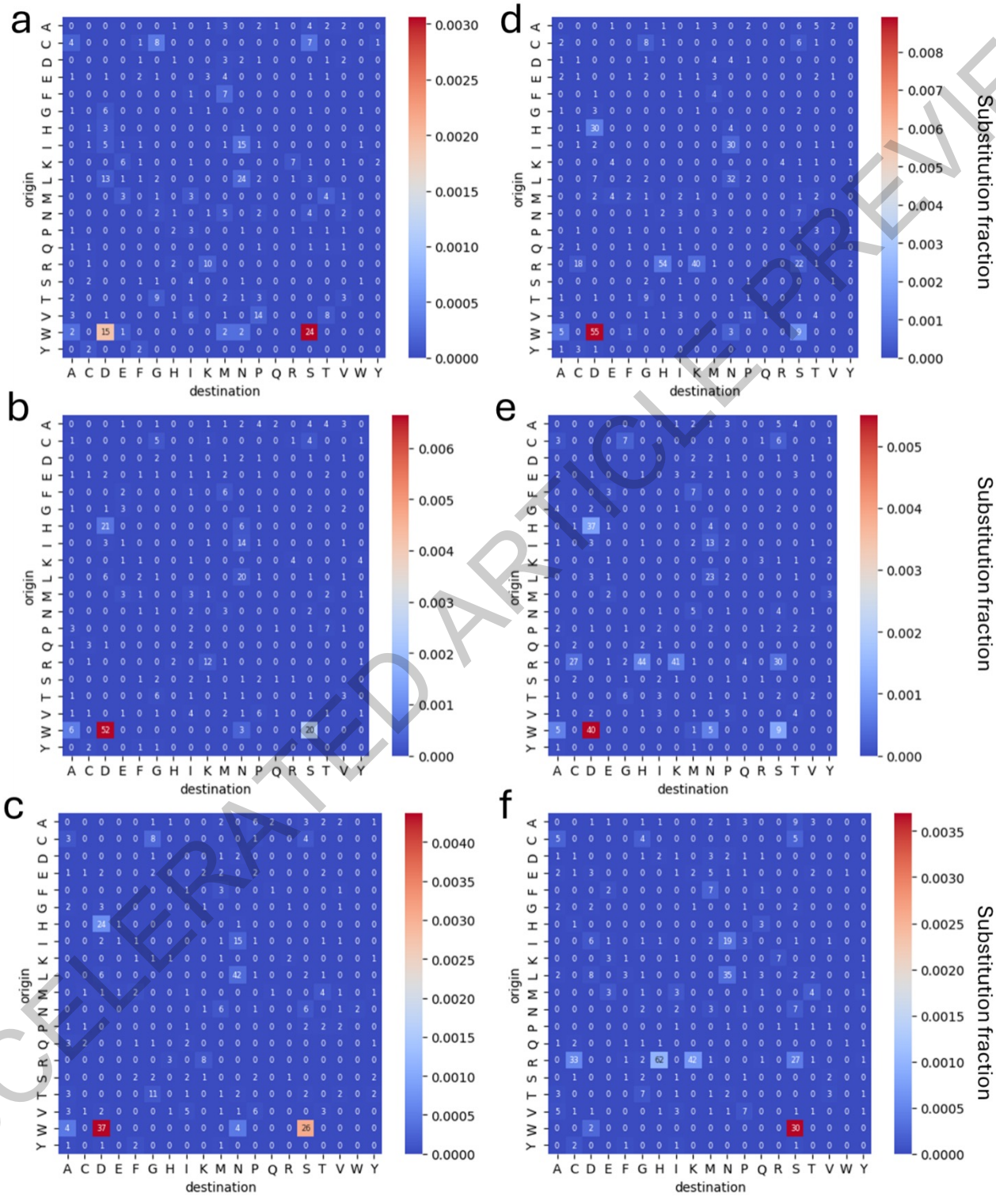


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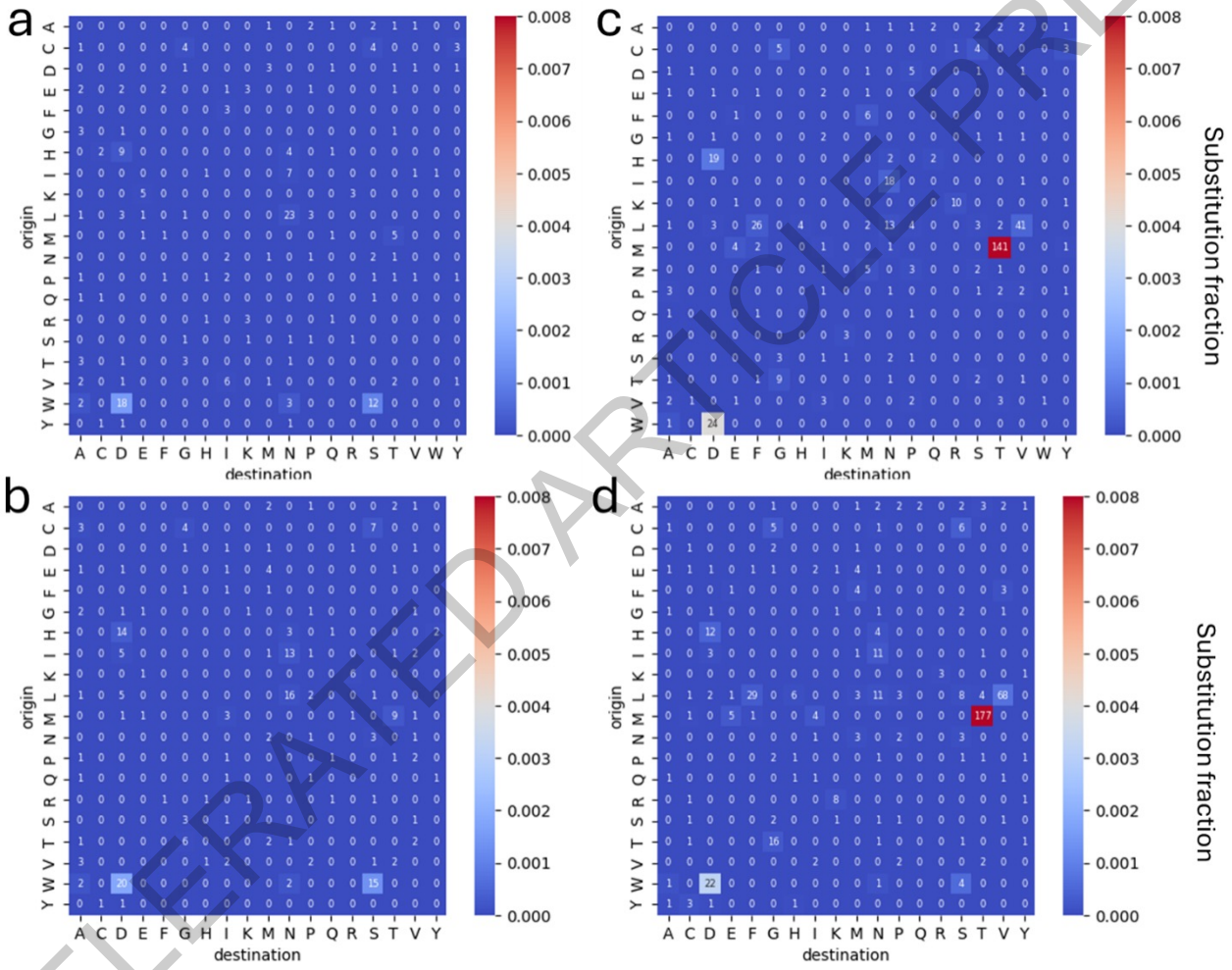


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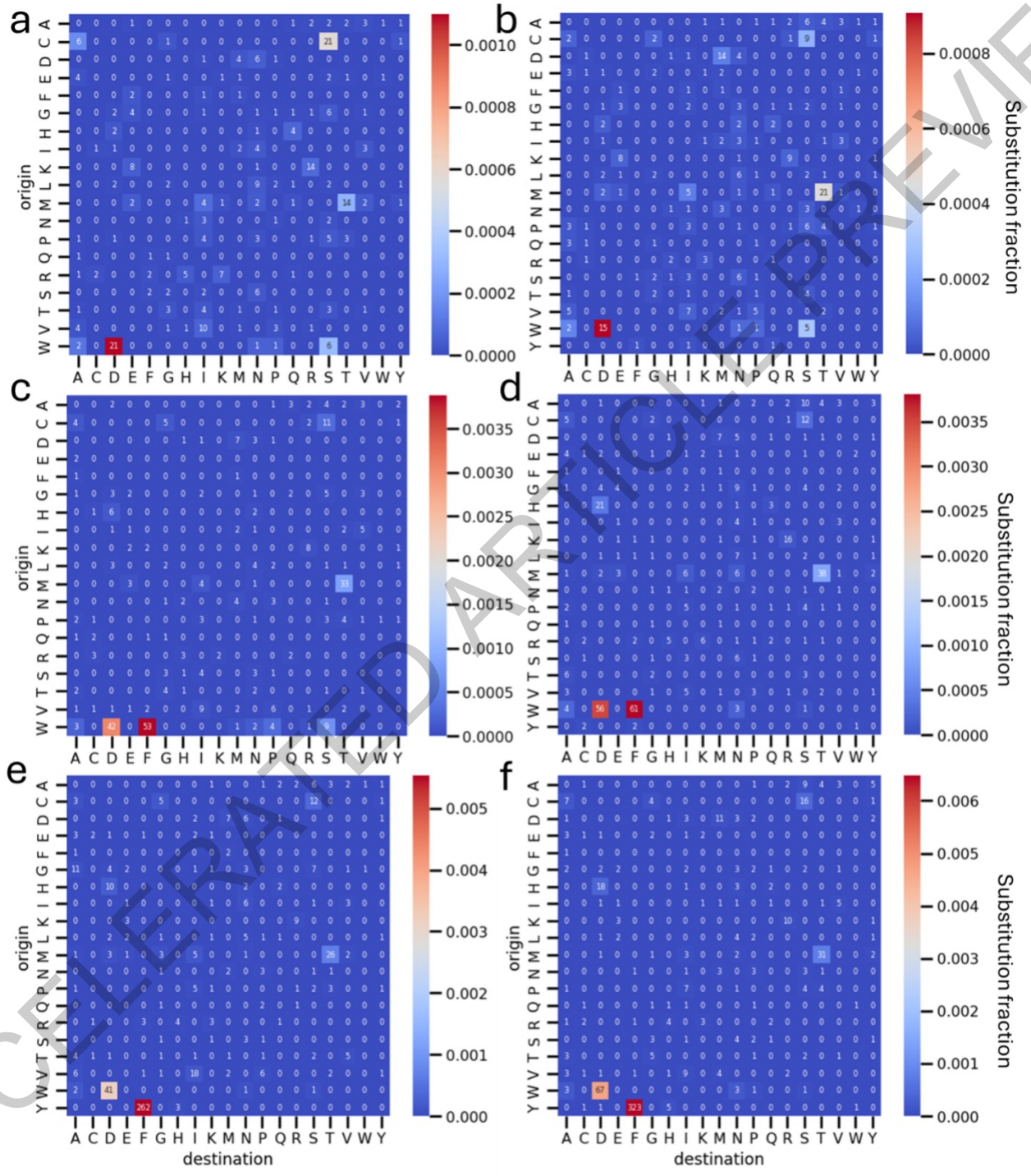


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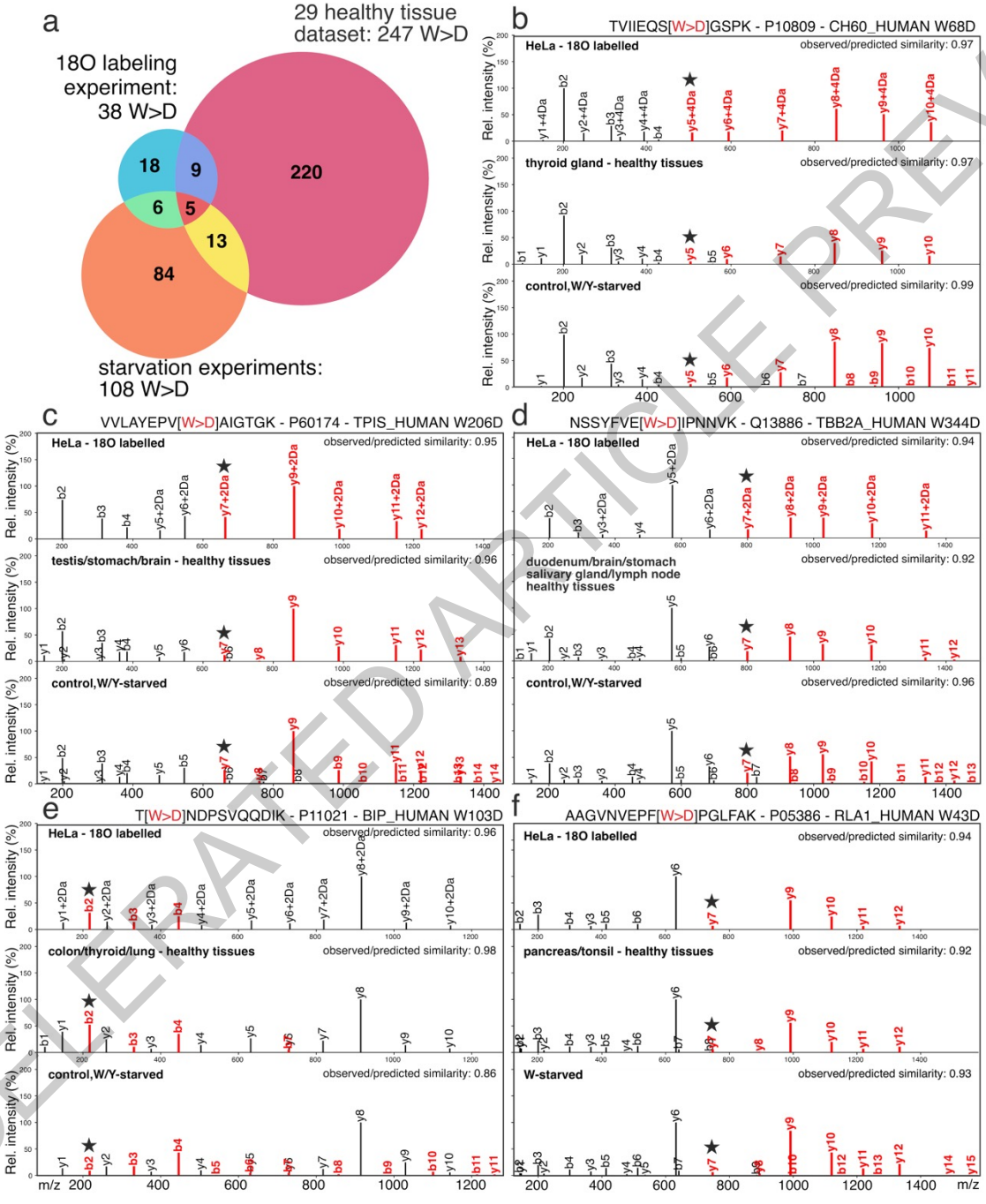


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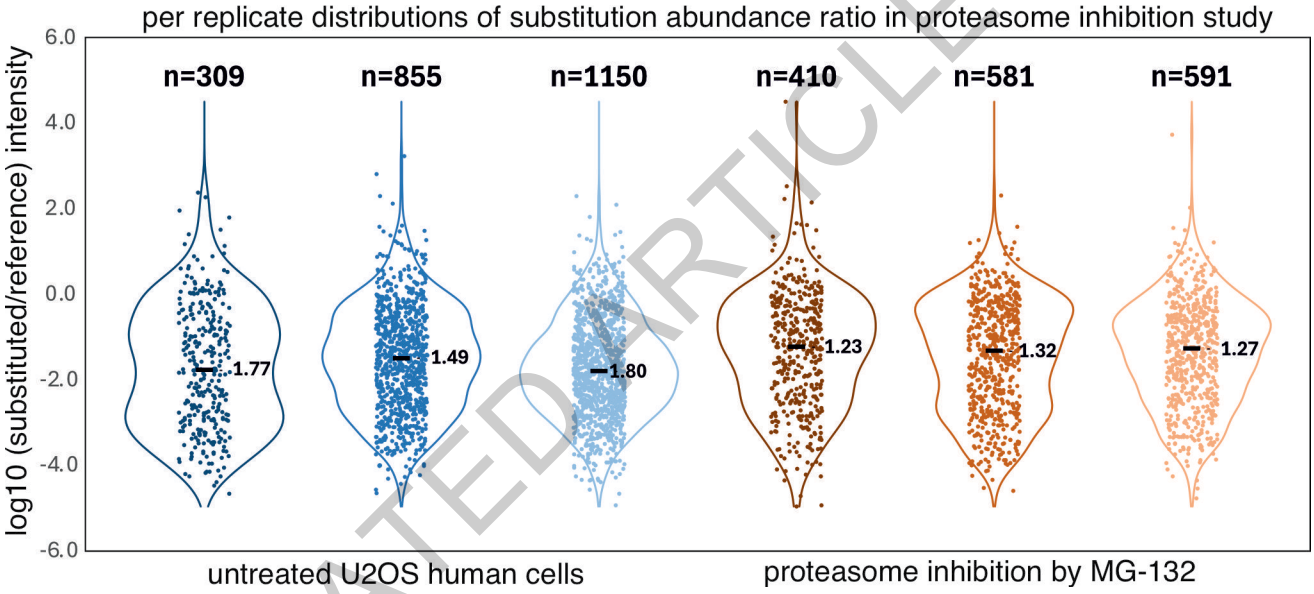


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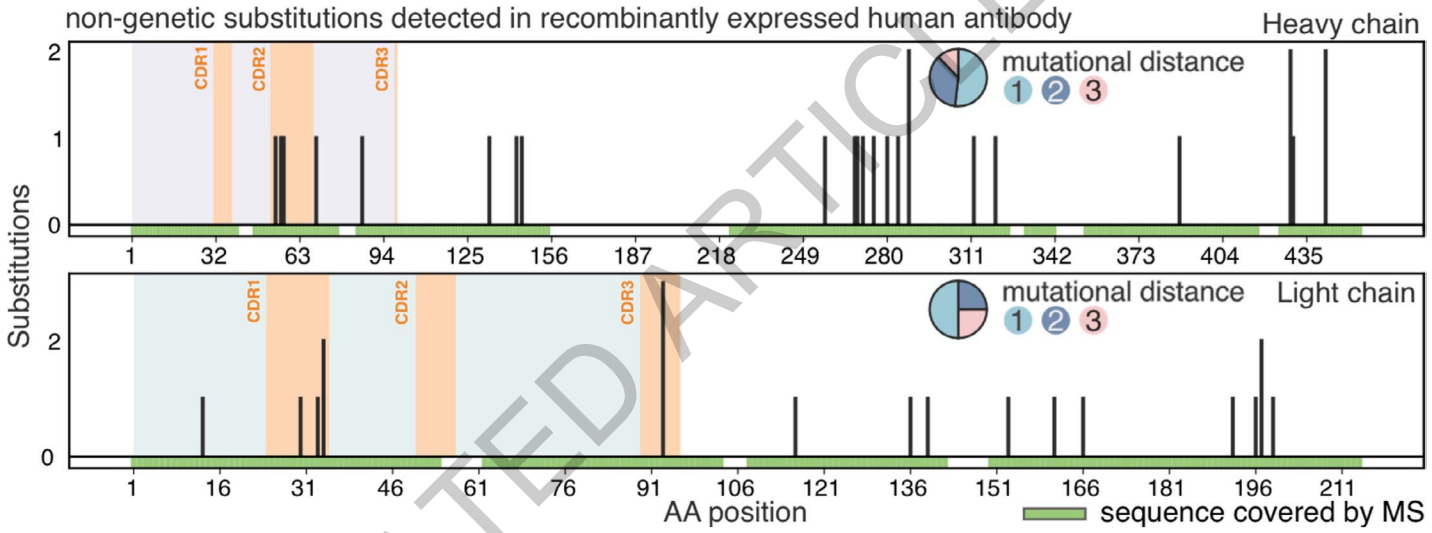


**Extended Data Fig. 10**

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|---|---|
| Data collection | no software used for data collection |
| Data analysis | https://github.com/treko90/tarsus<br>Fragpipe v.23.1<br>PepQuery 2.0.2<br>PDV 2.1.2<br>Python 3.11.5<br>MaxQuant 2.4.13.0<br>Byonic 5.10<br>Skyline 25.1<br>BWA 0.7.1<br>Samtools 1.22<br>pyAscore 1.0.1<br>IgBLAST - https://www.ncbi.nlm.nih.gov/igblast/<br>deTEL - https://git.mpi-cbg.de/tothpetroczylab/detelpy/-/tree/dfee20969a58e393274fe3d09625b43643381416/#multinomial-translation-error-landscape-mtel |

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

## Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Publicly available raw mass spectrometric datasets from https://www.proteomexchange.org/
PXD010154 healthy tissue data
PXD027328 degradomics dataset,
PXD017276 in vitro differentiated RBC precursors,
PXD043612 arginine starvation dataset,
PXD044085 leucine starvation dataset,
PXD028921 tryptophan/phenylalanine starvation dataset
PXD073135 datasets generated in this manuscript

Publicly available raw mass spectrometric datasets from https://massive.ucsd.edu/ProteoSAFe/static/massive.jsp
MSV000080679 BioPlex APMS

ensembl CDS sequences hg38, r112c- https://ftp.ensembl.org/pub/release-112/fasta/homo_sapiens/cds/
ensembl protein sequences hg38 r112 - https://ftp.ensembl.org/pub/release-112/fasta/homo_sapiens/pep/
AlphaMissense predictions - https://zenodo.org/records/8208688
GnomAD v4.1 - https://gnomad.broadinstitute.org/downloads

## Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

| | |
|---|---|
| Reporting on sex and gender | This information has not been collected |
| Reporting on race, ethnicity, or other socially relevant groupings | This information has not been collected |
| Population characteristics | This information has not been collected |
| Recruitment | No recruitment was done |
| Ethics oversight | No ethics oversight was performed |

Note that full information on the approval of the study protocol must also be provided in the manuscript.

# Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

# Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Sample size | For public data, we analyzed all available samples for each described experiment in this manuscript to obtain the most data. For experimental samples we ran one measurement per purified protein as the purpose of the analysis was to qualitatively confirm substitution existence without ambition to maximize the detections of all substitutions |
| Data exclusions | PXD010154 healthy tissue data - 27/29 tissues were analyzed; kidney and liver samples were not accompanied with clear RNAseq data<br>PXD027328 degradomics dataset - non fractionated samples were excluded due to low proteomic depth<br>PXD017276 in vitro differentiated RBC precursors - all RBC samples were analyzed<br>PXD043612 arginine starvation dataset - all fractionated samples were analyzed<br>PXD044085 leucine starvation dataset - all fractionated samples were analyzed<br>PXD028921 tryptophan/phenylalanine starvation dataset - all fractionated samples were analyzed |

| | |
|---|---|
| Replication | All experiments were performed in one replicate as the purpose of the analysis was to demonstrate the existence of substituted species and eventually to validate specific substitution types also identified in public data. No quantitative claims were raised. Confirmation of specific substitutions was performed using combination of our experimental datasets and publicly available data |
| Randomization | We did not perform randomization in our experiments as our purified protein samples had no comparison groups. |
| Blinding | We did not perform blinding as we focused on aggregated analysis of substitutions in humans without discrimination into groups |

# Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

## Materials & experimental systems

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ Antibodies |
| ☒ | ☐ Eukaryotic cell lines |
| ☒ | ☐ Palaeontology and archaeology |
| ☒ | ☐ Animals and other organisms |
| ☒ | ☐ Clinical data |
| ☒ | ☐ Dual use research of concern |
| ☒ | ☐ Plants |

## Methods

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ ChIP-seq |
| ☒ | ☐ Flow cytometry |
| ☒ | ☐ MRI-based neuroimaging |

## Plants

| | |
|---|---|
| Seed stocks | *Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.* |
| Novel plant genotypes | *Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.* |
| Authentication | *Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.* |